DESIGN-BUILD STATEMENT OF WORK

PROJECT NUMBER: EEPZ252131

CLIN 001 - RENOVATE/CONVERT 50th 49th SQ OPS, B230 ***(Base Bid)***

CLIN 002 – CONVERT CLASSROOM 04 TO OPEN STORAGE AREA, B230 ***(Bid Option #1)***

COLUMBUS AFB, MISSISSIPPI

20 APRIL 2026

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CHAPTER 1 NOMENCLATURE

1.1 ACRONYMS & ABBREVIATIONS

AF	Air Force
AFI	Air Force Instruction
AFM	Air Force Manual
AFUE	Annual Fuel Utilization Efficiency
AHU	Air Handling Unit
ATFP	Anti-terrorism/Force Protection
BCE	Base Civil Engineer
CAD	Computer Assisted (or Aided) Design
CAFB	Columbus Air Force Base
CDs	Construction Documents
CEPM	Civil Engineering Project Manager
CES	Civil Engineering Squadron
cfm	cubic feet per minute
CFR	Code of Federal Regulations
CHW	Chilled Water
CMI	14CES Construction Manager Inspector
CO	Contracting Officer
COR	Contracting Officer Representative
Comm	Communications
CPM	Contractor's Project Manager
CSS	Contractor's Site Superintendent
DP	Design Professional
EMCS	Energy Management & Control System
EMI	Electromagnetic Interference
EPA	Environmental Protection Agency
ETL	Engineering Technical Letter
FAR	Federal Acquisition Regulation
FCU	Fan Coil Unit
FM	Factory Mutual
FOD	Foreign Object Debris
fps	feet per second
ft.	foot or feet
GFCI	Government Furnished Contractor Installed
gpm	gallons per minute
Hazmat	Hazardous Materials
HMW	Hazardous Material or Waste
HTW	Hazardous and Toxic Waste
HVAC	Heating, Ventilating and Air Conditioning
NEC	National Electrical Code
NTP	Notice to Proceed
O&M	Operation and Maintenance
OSP	Originally Specified Product
PDF	Portable Document Format
PE	Professional Engineer
Ph	Phase (electrical)
PM	Project Manager
psi	pounds per square inch
QCP	Quality Control Plan
RFP	Request for Proposal
SOW	Statement of Work
TI	Technical Instruction
TL	Technical Letter
TM	Technical Manual
UFC	Unified Facilities Criteria
UPS	Uninterruptible Power Supply
VAV	Variable Air Volume
V	Volts
W	Watts

1.2 BASIC TERMS

The following table provides many of the meanings of words or phrases commonly used in this document.

<table>
<tr><td colspan="2">All directives, informative, etc. statements are directed to the Contractor even if the term “contractor” or the contractor name is not included (e.g. “Provide a nail.” is equivalent to “The contractor shall provide a nail.” And “remove” is equivalent to “The Contractor shall remove”). Also, the Contractor shall ably and competently execute all directives including all directly or indirectly related tasks required to safely and legally perform the directive (e.g. “remove debris from CAFB” includes “safe and legal removal, disposal and obtaining all necessary permits, etc.)</td></tr>
<tr><td colspan="2">A device, equipment or system referred to in the singular (e.g. “the pump”, or “the ___ system”) includes all related devices, elements and components required to Provide (see below in this table) that device, equipment, or system.</td></tr>
<tr><td>The Term</td><td>Shall Mean (*=Shall in Addition Mean)</td></tr>
<tr><td>Approved (by CAFB)</td><td>*without waiving or suspending the full force of Contractor obligation to fulfill the CDs</td></tr>
<tr><td>Aspect</td><td>Distinct feature or element of the project</td></tr>
<tr><td>CAFB</td><td>Government contracting Authority on Columbus Air Force Base</td></tr>
<tr><td></td><td></td></tr>
<tr><td>Construction Documents</td><td>Drawings, specifications and their references including all Design Criteria and Design & Construction Requirements for this Project.</td></tr>
<tr><td>Contractor</td><td>The Design-Build entity holding a current Government contract including all personnel in direct or indirect employ of same entity servicing same contract including partners, consultants, subcontractors, suppliers, and manufacturers.</td></tr>
<tr><td>Design Professional</td><td>(Abbreviated: DP) Professionals including Architects, Engineers and Landscape Architects licensed to practice in one or more states of the U.S.</td></tr>
<tr><td>Day</td><td>Calendar day unless noted otherwise.</td></tr>
<tr><td>DP</td><td>Licensed Design Professional fully responsible for all professional design on this project.</td></tr>
<tr><td></td><td>Engineer DP</td></tr>
<tr><td>For example (e.g.)</td><td>One among many possible examples</td></tr>
<tr><td>Government</td><td>Columbus Air Force Base Contracting Officer</td></tr>
<tr><td>Including [Includes]</td><td>Including, but not limited to [Includes, but is not limited to]</td></tr>
<tr><td>Provide</td><td>Furnish and install complete, functional, and ready for use without added government action</td></tr>
<tr><td>Provision</td><td>Furnishing & installation complete, functional, and ready for use without added government action</td></tr>
<tr><td>Will, must, shall</td><td>& similar Terms are mandatory directives whereby ‘Term’ = ‘Term + to CAFB satisfaction’</td></tr>
<tr><td>Work</td><td>The whole or any part, product, system or subsystem of this project</td></tr>
<tr><td>Replace</td><td>Demolish and dispose of existing element. Provide new element of equal or greater value.</td></tr>
<tr><td>Review</td><td>Review and, if and when acceptable to the government, approval</td></tr>
<tr><td>Subcontractor</td><td>Subcontractor and all personnel in Subcontractor’s direct or indirect employ on any given Project including suppliers and manufacturers</td></tr>
</table>

CHAPTER 2 DESCRIPTION OF PROJECT

2.1 SUMMARY

CLIN 001 – RENOVATE/CONVERT 50th 49th SQ OPS, B230 ***(Base Bid)***

- The Contractor must provide all plant, labor, equipment, transportation, and tools required to complete the renovation of B230 to be converted into a consolidated Flying Training Squadron (FTS) facility. Work includes replacing interior finishes, upgrading electrical fixtures, demolishing and erecting interior partition walls and doors, mechanical and plumbing work, among others. Perform all work according to this Statement of Work (SOW), the Concept Design Drawings, and all related AF requirements and other codes and references.

CLIN 002 – CONVERT CLASSROOM 04 TO OPEN STORAGE AREA, B230 ***(Bid Option #1)***

- In addition to the base bid, this bid option requires additional work in Classroom 4 to convert this space into an Open Storage Area (OSA) for classified information. Work includes the installation of an Intrusion Detection System (Honeywell Vindicator), including cameras, motion sensors, magnetic door switch, and access control system for CAC card access. Work also includes hardening/ securing the room perimeter, installing a dedicated electric panel from the MDP, and installing SIPR communications drops.

2.2 BACKGROUND

Columbus AFB is one of three Specialized Undergraduate Pilot Training (SUPT) bases and one of three bases to perform Introduction to Fighter Fundamentals (IFF) for prospective fighter pilots. B230 is currently utilized for classroom and briefing space by the 48th FTS. B230 will soon be utilized by the 49th & 50th FTS, after divestment of the T-1 Jayhawk in Spring 2026, in preparation for the upcoming T-7A Red Hawk mission. The consolidation of TFS into B230 is essential to the flying training mission supporting 16,000+ sorties per year as part of Columbus AFB SUPT and IFF syllabus.

2.3 SCOPE OF PROJECT

<u>CLIN 001 - RENOVATE/CONVERT 50th 49th SQ OPS, B230 ***(Base Bid)***</u>

2.3.1 Demolish existing flooring and cove base, including all adhesive. Repair floor and wall surfaces as required to prepare for installation of new flooring and cove base. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165, Fitness Room + 2 Adjacent Rooms, Step Desk.

2.3.2 Furnish and install new 20 mil, 5mm thick, LVT plank flooring and associated installation materials. Furnish and install new transition strips at locations with differing floor types. LVT style and color shall match Shaw Contract's Inlet Shade 26506 or approved equal. Confirm style and color match with the Base Architect. Applies to the following spaces:

- Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37 (Mother Room Portion), Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Step Desk.

2.3.3 Furnish and install new carpet flooring and associated installation materials. Furnish and install new transition strips at locations with differing floor types. Carpet style and color shall match Shaw Contract's Collective V Tile Charcoal 38500 or approved equal. Confirm style and color match with the Base Architect. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Fitness Room + 2 Adjacent Rooms.

2.3.4 Furnish and install new concrete solid stain, non-slip epoxy, and associated installation materials. Furnish and install new transition strips at locations with differing floor types. Concrete solid stain color shall match the color of Shaw Contract's Inlet Shade 26506 or approved equal. Epoxy shall be low shine and clear, having no color. Confirm styles and colors with the Base Architect. Applies to the following spaces:

- Rooms 40/41.

2.3.5 Furnish and install new 4" vinyl wall base. Confirm style and color match with the Base Architect.

- Vinyl wall base style and color shall match Johnsonite's 20 Charcoal WG or approved equal. Applies to the following spaces:
    - Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37 (Mother Room Portion), Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Step Desk.
- Vinyl wall base style and color shall match Johnsonite's 40 Black B or approved equal. Applies to the following spaces:
    - Classroom 1, Classroom 2, Classroom 3, Fitness Room + 2 Adjacent Rooms.

2.3.6 Demolish the existing suspended ceiling grid system and tiles. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165, Fitness Room + 2 Adjacent Rooms, Step Desk.

2.3.7 Furnish and install new tegular 2'-0" x 2'-0" suspended ceiling grid system and ceiling tiles. The new suspended ceiling shall maintain existing ceiling height. Grid system and tiles shall match and be white. Color to be approved by the Base Architect. Tiles are not to be vinyl coated. Applies to the following spaces:

- 5/8" acoustical ceiling tiles required: Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37 (Mother Room Portion), Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.
- 5/8" moisture resistant ceiling tiles required: Room 37 (Men's Room Extension).

2.3.8 Furnish and install all materials necessary to repair the existing interior walls. Areas requiring repair include, but are not limited to, water and moisture damage, holes, peels/cracks/chips, among others. Repair walls with materials that match existing. All repairs shall have an uninterrupted, flat wall surface transition with existing. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.

2.3.9 Paint new and existing interior walls. Paint color(s) shall be Sherwin Williams' Accessible Beige or approved equal. Color should conform to base standards and be approved by the Base Architect. Interior walls will be painted to match the color selected by the Base Architect. At a minimum, painting should include one (1) coat of primer, two (2) coats of paint, sanding and prepping of all surfaces, and placing drop cloths on the floor. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37 (Mothers Room Portion), Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.

2.3.10 Repair or replace disturbed building components impacted by construction actions of this project, match existing.

2.3.11 Any repaired or replaced wall surfaces shall achieve level 4 finish.

2.3.12 All door hardware must be Best 7-pin core compatible.

2.3.13 **Classroom 1** Specific Architectural Work:

- Demolish the existing interior partition wall protruding into the classroom from the south end. Repair the adjacent walls and floor with materials that match existing.

2.3.14 **Classroom 2** Specific Architectural Work:

- Demolish the interior door, doorframe, and associated hardware of the briefing room.
- Demolish the existing interior partition walls separating Classroom 2 and the briefing room. Repair the adjacent walls and floor with materials that match existing.
- Furnish and install all materials required to construct the new interior partition walls for the new briefing room. The new briefing room shall be located at the opposite end of Classroom 2 to match Classrooms 1 and 3, see concept drawing for location. Materials include, but are not limited to, metal framing, insulation, 5/8" drywall, wall surface preparation materials, and paint, to achieve a level 4 finish. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.
- Furnish and install one (1) new 3'x7' opaque solid core wood interior door that meets current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for the new briefing room. Locks and hinges should be mortise. The new briefing room door shall be located to match Classrooms 1 and 3 doors, see concept drawing for location. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.

2.3.15 **Room 5** Specific Architectural Work:

- Demolish the interior door, doorframe, and associated hardware of the storage room.
- Demolish the existing interior partition wall separating the storage room and student lounge. Repair the adjacent walls and floor with materials that match existing.

2.3.16 **Room 11** Specific Architectural Work:

- Demolish all interior doors, doorframes, and associated hardware. Repair the walls and floor as necessary to prepare for one (1) cased opening and infill of three (3) existing doors. See concept drawing for locations. Materials used for repairs shall match existing.

- Furnish and install all materials required for the 3’x7’ cased opening in the existing door opening. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install all materials required to infill the three (3) existing door openings to match existing wall assemblies. Repaired wall surfaces shall achieve level 4 finish.
- Demolish an area centered on the wall between existing Room #11 and Lobby to prepare for one (1) 4’x7’ cased opening. See concept drawing for location. Consult with a licensed structural engineer prior to demolition. Repair the walls and floor as necessary. Materials used for repairs shall match existing. Repaired wall surfaces shall achieve level 4 finish. Furnish and install all materials required for the cased opening.
- Furnish and install all materials required to construct the new interior partition walls for ten (10) two-ship briefing rooms, one (1) storage closet, and new corridor. Materials include, but are not limited to, metal framing, insulation, 5/8” drywall, wall surface preparation materials, and paint, to achieve a level 4 finish. The new corridor shall maintain 4’ width throughout, be centered on the cased openings, and comply with egress requirements. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.
- Furnish and install all new 3’x7’ opaque solid core wood interior doors that meet current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for the two-ship briefing rooms and storage closet. Provide louvered panels at the bottoms of the 10 briefing rooms doors for a return air path. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.

2.3.17 **Room 17** Specific Architectural Work:

- Demolish the interior door, doorframe, and associated hardware of Room 17.
- Demolish the existing interior partition walls on Room 17. Repair the adjacent walls and floor with materials that match existing.
- Furnish and install all materials required to construct the new interior partition walls for three (3) offices. Materials include, but are not limited to, metal framing, insulation, 5/8” drywall, wall surface preparation materials, and paint, to achieve a level 4 finish. The new corridor outside these offices shall maintain 8’ width throughout and comply with egress requirements. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.
- Furnish and install all one (1) new 3’x7’ opaque solid core wood interior door for each office that meet current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.

2.3.18 **Room 36** Specific Architectural Work:

- Demolish existing sink. Cap hot and cold-water lines to prepare for installation of one (1) new sink.
- Demolish existing base cabinets and shelving, countertops, wall mounted counters, and associated components.
- Furnish and install one (1) new stainless-steel undermount double compartment sink 10” deep with a stainless-steel manual, two handle faucet. Reuse existing plumbing components including rough-ins, stubs, hot and cold-water lines, among others to the maximum extent possible. Furnish and install all new plumbing components as required to supply the room with a complete and functional plumbing system. Repair or replace disturbed wall and floor surfaces as result of construction actions from this project, match to existing. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install one (1) new recessed washer box with cold- and hot-water lines, shut-off valves for each water line, waste lines, and associated components required for fixture connection hookups. Reuse existing plumbing components rough-ins, stubs, hot and cold-water lines, among others to the maximum extent

possible. Furnish and install all new plumbing components as required to supply the room with a complete and functional plumbing system. Repair or replace disturbed building components impacted by construction actions of this project, match to existing. Repaired wall surfaces shall achieve level 4 finish.

- Furnish and install one (1) dryer vent ducted to the outside with wall or ceiling cap and associated components. Furnish and install all new components as required to supply the room with a complete and functional dryer vent system. Repair or replace disturbed building components impacted by construction actions of this project, match to existing. Repaired wall surfaces shall achieve level 4 finish.
- Design, furnish and install all materials necessary to construct a new raised wood L-shape counter and linear counter, see concept drawing for locations. This includes, but is not limited to, 36” high x 24” deep base cabinets, doors, adjustable shelves, drawers, solid surface countertops with 4” backsplash, knobs, handles, soft-closing mechanisms, hardware, fasteners, and anything else required for complete and functional Life Support counters. The L-shape counter shall be designed with three (3) workstations. The counters and associated materials shall be prefinished by the manufacturer, including color. The counters shall be stained to match Sherwin-Williams’ Walnut MW439 or approved equal, with semi-gloss finish. The countertops shall be Bellavati’s Coppermill DFS1-404 or approved equal. Styles and colors shall be approved by the Base Architect.
- Design, furnish and install all materials necessary to construct wall cabinets mounted above the L-shape counter and linear counter, see concept drawing for locations. This includes, but is not limited to, 36” high x 12” deep wall cabinets, doors, adjustable shelves, knobs, handles, soft-closing mechanisms, concealed hinges, hardware, fasteners, and anything else required for complete and functional Life Support cabinets. The cabinets and associated materials shall be prefinished by the manufacturer, including color. The cabinets shall be stained to match Sherwin-Williams’ Walnut MW439 or approved equal, with semi-gloss finish. Styles and colors shall be approved by the Base Architect.
- Demolish existing wainscotting, wall panels, wall mounted shelves, and associated components. Repair walls with materials that match existing. All repairs shall have an uninterrupted, flat wall surface transition with existing.
- Demolish area on the wall in the bottom right corner of room 36 to prepare for one (1) 6’x7’ double door and doorframe. See concept drawing for locations. Consult with a licensed structural engineer prior to demolition. Repair the walls and floor as necessary. Materials used for repairs shall match existing. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install one (1) 6’x7’ opaque solid core wood interior door that meets current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for the new briefing room. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.

2.3.19 **Room 37 (Mothers Room Portion)** Specific Architectural Work:

- Furnish and install all materials required to construct the new interior partition wall that separates the bathroom extension and Mother’s Room. See concept drawing for location. Materials include, but are not limited to, metal framing, insulation, 5/8” gypsum wallboard, 5/8” cementitious backerboard where wall tile is specified, wall surface preparation materials, and paint, to achieve a level 4 finish. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.

2.3.20 **Room 43** Specific Architectural Work:

- Demolish all interior doors, doorframes, and associated hardware. Repair the walls and floor as necessary to prepare for one (1) cased opening and infill of two (2) existing doors. See concept drawing for locations. Materials used for repairs shall match existing.
- Furnish and install all materials required for the 3’x7’ cased opening in the existing door opening. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install all materials required to infill the two (2) existing door openings to match existing wall assemblies. Repaired wall surfaces shall achieve level 4 finish.

- Furnish and install all materials required to construct the new interior partition walls for two (2) four-ship briefing rooms, two (2) flex-ship briefing rooms capable of seating eight occupants, one (1) storage closet, and new corridor. Materials include, but are not limited to, metal framing, insulation, 5/8" drywall, wall surface preparation materials, and paint, to achieve a level 4 finish. The new corridor shall maintain 4' width throughout, be centered on the cased opening, and comply with egress requirements. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.
- Furnish and install all new 3'x7' opaque solid core wood interior doors that meet current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for the two-ship briefing rooms and storage closet. Provide louvered panels at the bottoms of the briefing room doors for a return air path. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.

2.3.21 **Rooms 156/157/158/159** Specific Architectural Work:

- Demolish the interior double door, doorframe and associated hardware of Room 156. Repair the walls and floor as necessary to prepare for one (1) 6'x7' double door and doorframe to be installed. See concept drawing for locations. Materials used for repairs shall match existing. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install all new 6'x7' opaque solid core wood interior double door that meet current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for existing Room 156. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.
- Demolish all the interior doors, doorframes, and associated hardware of Rooms 157 and 158. Repair the walls and floor as necessary to prepare for the infill of the existing doors. See concept drawing for locations. Materials used for repairs shall match existing.
- Furnish and install all materials required to infill the existing door openings of Rooms 157 and 158 to match existing wall assemblies. Repaired wall surfaces shall achieve level 4 finish.
- Demolish the interior door, doorframe and associated hardware of Room #159. Demolish an area on the wall next to the door to prepare for one (1) 6'x7' double door and doorframe to be installed. See concept drawing for location. Consult with a licensed structural engineer prior to demolition. Materials used for repairs shall match existing. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install all new 6'x7' opaque solid core wood interior double door that meet current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for existing Room 159. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.
- Demolish the existing interior partition walls to create the Equipment Locker Room. Repair the adjacent walls and floor with materials that match existing. Consult with a licensed structural engineer prior to demolition.

2.3.22 **Rooms 165** Specific Architectural Work:

- Furnish and install all materials required to construct a new interior partition wall to create a new small office (Room 165 NCOIC Office). The new office shall be located at the bottom left corner of the space, utilizing the existing hallway entrance door, see concept drawing for location. Materials include, but are not limited to, metal framing, insulation, 5/8" drywall, wall surface preparation materials, and paint, to achieve a level 4 finish. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.
- Demolish area on the wall in the top right corner of room 165 to prepare for one (1) 3'x7' door and doorframe. See concept drawing for locations. Consult with a licensed structural engineer prior to

demolition. Repair the walls and floor as necessary. Materials used for repairs shall match existing. Repaired wall surfaces shall achieve level 4 finish.

- Furnish and install one (1) new 3’x7’ opaque solid core wood interior door that meets current ASHRAE standards, interior knock-down doorframes, associated door hardware including hinges, handles, and locksets (coordinate with Base Locksmith for requirements), kickplates, and any other necessary components for the new briefing room. Locks and hinges should be mortise. Materials shall match with others throughout the unit for uniformity. Materials shall be prefinished by the manufacturer, including color, and be approved by the Base Architect.

2.3.23 **Fitness Room + 2 Adjacent Rooms** Specific Architectural Work:

- Demolish the interior doors, doorframes, and associated hardware of both the center room and rightmost room. Repair the walls and floor as necessary to prepare for infill of the existing doors. Repaired wall surfaces shall achieve level 4 finish.
- Demolish the existing interior partition walls to create Classroom 4. Repair the adjacent walls and floor with materials that match existing. Consult with a licensed structural engineer prior to demolition.

2.3.24 **Step Desk** Specific Architectural Work:

- Demolish existing base cabinets, countertops, wall mounted counters, and associated components.
- Demolish existing raised floor and all associated components.
- Design, furnish and install all materials necessary to construct a new raised wood L-shape desk in the Step Desk area, see concept drawing for location. This includes, but is not limited to, 34” high x 24” deep base cabinets, doors, adjustable shelves, drawers, solid surface countertops, knobs, handles, soft-closing mechanisms, concealed hinges, hardware, fasteners, and anything else required for a complete and functional step desk. The step desk shall be designed with four (4) workstations, have a standing height countertop no shorter than 4’ AFF run the entire length of the desk, and a section that complies with current ABA standards. The sides of the step desk that face the corridor shall be flat surface with no visible joints. The corridor around the step desk shall maintain 8’ width throughout and comply with egress requirements. The step desk and associated materials shall be prefinished by the manufacturer, including color. The step desk shall be stained to match Sherwin-Williams’ Walnut MW439, or approved equal, with semi-gloss finish. The countertops shall be Bellavati’s Coppermill DFS1-404, or approved equal. Styles and colors shall be approved by the Base Architect.
- Design, furnish and install all materials necessary to construct a new raised wood wall mounted counter in the Step Desk area, see concept drawing for location. This includes, but is not limited to, 18” high x 24” deep wall mounted counters, doors, adjustable shelves, solid surface countertop, knobs, handles, soft-closing mechanisms, concealed hinges, hardware, fasteners, and anything else required for a complete and functional counter. The counter shall be designed with four (4) workstations and have a standing height countertop no shorter than 4’ AFF. The counter and associated materials shall be prefinished by the manufacturer, including color. The counter shall be stained to match Sherwin-Williams’ Walnut MW439, or approved equal, with semi-gloss finish. The countertops shall be Bellavati’s Coppermill DFS1-404, or approved equal. Styles and colors shall be approved by the Base Architect.
- Demolish the interior door, doorframe, and associated hardware between existing corridor to Room #17 and the step desk.
- Demolish the existing interior partition wall separating existing Room #17 and the step desk. Consult with a licensed structural engineer prior to demolition. Repair the adjacent walls and floor with materials that match existing.

2.3.25 **Men’s Restroom North** Specific Architectural/Plumbing Work:

- Remove all components of the existing bathroom accessories and mirrors. Protect and store all components during demolition. Reinstall all components.
- Keep the existing floor drain, waste piping, and associated components in place.
- Demolish the existing lavatory, water closet, and urinals. All existing carriers shall be removed.

- Remove all water lines back to their main and cap. Remove all waste piping back to their main and cap. Repair or replace disturbed building components impacted by construction actions of this project, match to existing. Repaired wall surfaces shall achieve level 4 finish.
- Demolish the existing stall partitions in their entirety.
- Demolish the existing porcelain wall tile and cementitious backerboard.
- Demolish the existing porcelain floor tile to prepare for new flooring. Prepare the floors as required, including removing existing glue adhesive, for installation of new porcelain floor tile.
- Demolish the existing lavatory wet wall to prepare for future lockers. Repair or replace disturbed building components impacted by construction actions of this project, match to existing. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install one (1) new shower with associated plumbing fixtures and components. Shower shall be of a white one-piece 36” square shower stall with shower kit including controls and fittings, as well as a centered drain. Shower shall have two (2) corner shelves, at a minimum. Shower head shall be ultralow flow (1.5 GPM). Contractor shall ensure the shower base is slip-resistant and has adequate slope to direct water towards the drain. Furnish and install all new piping and associated components necessary to connect the shower to existing hot- and cold-water and waste lines. Water lines shall be PP-RCT, manufacturer to be Aquatherm or approved equal. Water lines shall connect to new thermostatic mixing (anti-scald) valves. Furnish and install wall shut off valves for the shower’s water lines.
- Furnish and install new stainless steel grab bars in the shower stall. Components shall be installed using best practices for location placement without hindering user mobility.
- Furnish and install one (1) new floor-mounted flush valve water closet with associated plumbing fixtures and components. Water closet shall be Zurn brand or approved equal. Flush handle shall be silver in color. Furnish and install all new carriers and associated components. Furnish and install all new piping and associated components necessary to connect the water closets to existing water and waste lines. Water lines shall be PP-RCT, manufacturer to be Aquatherm or approved equal. Furnish and install wall shut off valves for the water closet’s water lines.
- Furnish and install a new wall-mounted open industrial bathroom vanity with one (1) white drop-in lavatory and associated plumbing fixtures and components. Vanity shall be Corian Quartz’s Antique Pearl or approved equal solid surface countertop with 4” integrated backsplash. Vanity shall be prefinished by the manufacturer, including color. Lavatory shall be KOHLER brand or approved equal. Faucet shall be Delta brand or approved equal, have two (2) handles for hot- and cold-water operation, and be silver in color. Furnish and install all new piping and associated components necessary to connect the lavatories to existing water and waste lines. Furnish and install wall shut-off valves for each lavatory’s water lines.
- Furnish and install all materials required to construct the new interior partition wall between the shower stall and water closet. Materials include, but are not limited to, metal framing, insulation, 5/8” cementitious backerboard, wall surface preparation materials, and paint, to achieve a level 4 finish. Design, furnish, and erect the new interior partition wall in accordance with NFPA 101.
- Furnish and install new 12” x 12” porcelain floor tile and associated installation materials. Furnish and install new transition strips at locations with differing floor types. The tile shall match Daltile’s Volume 1.0 Thunder VL62 or approved equal. Confirm color and dimensions match with the Base Architect. Grout color shall match Bostik Delorean Gray H160 or approved equal. Furnish and install new Schluter DILEX-EHK cove-shaped profile or approved equal for inside wall corners and floor/wall transitions. Transitions shall be brushed stainless steel, confirm color with the Base Architect
- Furnish and install new 5/8” cementitious backerboard where wall tile is specified, furnish mounting/fastening hardware for installation. Furnish all materials necessary to prepare the cementitious backerboard as required for installation of new wall tiles.
- Furnish and install from the floor to 6’ high new 12” x 12” porcelain wall tile and associated installation materials. The tile shall match Daltile’s Volume 1.0 Degrees Silver VL71 or approved equal. Confirm color and dimensions match with the Base Architect. Grout color shall match Bostik Delorean Gray H160 or approved equal. Furnish and install new Schluter DILEX-EHK cove-shaped profile or approved equal for inside wall corners and floor/wall transitions. Transitions shall be brushed stainless steel, confirm color with the Base Architect.

- Furnish and install new 5/8” GWB where cementitious backerboard is not, furnish mounting/fastening hardware for installation.
- Furnish all materials necessary to prepare the gypsum wall boards to achieve a level 4 finish. Ensure the newly surfaced walls are installed to have an uninterrupted, flat wall surface.
- Furnish and install all materials necessary for painting. Paint all walls, except where new wall tile is to be installed, to match color to Sherwin-Williams’ Accessible Beige SW7036 or approved equal. Confirm color with the Base Architect. At a minimum, painting should include one (1) coat of primer, two (2) coats of paint with satin/egg-shell finish, sanding and prepping of all surfaces, and placing drop cloths on the floor.
- Furnish and install new shower and stall partitions, associated hardware, and mounting components. These stalls do not need to achieve ABA compliance. Color of the stall partitions shall be prefinished by the manufacturer and approved by the Base Architect.
- Any damage to the new floor and wall tiles caused by the Contractor while mounting stall partitions and bathroom and plumbing fixtures and accessories will be fixed or replaced by the Contractor, at the Contractor’s expense.

2.3.26 **Room 37 (Men’s Restroom Extension)** Specific Architectural/Plumbing Work:

- Remove the recessed paper towel dispenser and surface mounted waste receptacles from the men’s restroom. Reinstall the components on wall located between the bathroom door and lavatory. Repair or replace disturbed building components impacted by construction actions of this project, match to existing. Repaired wall surfaces shall achieve level 4 finish.
- Demolish an area centered on the wall between existing Room 37 and Men’s Restroom to prepare for one (1) 4’x7’ cased opening. See concept drawing for location. Consult with a licensed structural engineer prior to demolition. Repair the walls and floor as necessary. Materials used for repairs shall match existing. Repaired wall surfaces shall achieve level 4 finish.
- Furnish and install all materials required for the 3’x7’ cased opening in the existing door opening.
- Furnish and install all materials required to construct the new interior partition walls for the wet wall. See concept drawing for location. Materials include, but are not limited to, metal framing, insulation, 5/8” cementitious backerboard where wall tile is specified, wall surface preparation materials, and paint, to achieve a level 4 finish. Design, furnish, and erect the new interior partition walls in accordance with NFPA 101.
- Furnish and install one (1) new floor drain, waste piping, and associated components in the bathroom extension. Repair or replace disturbed building components impacted by construction actions of this project, match to existing.
- Furnish and install two (2) new showers with associated plumbing fixtures and components. Showers shall be of a white one-piece 36” square shower stall with shower kit including controls and fittings, as well as a centered drain. Showers shall have two (2) corner shelves each, at a minimum. Shower heads shall be ultralow flow (1.5 GPM). Contractor shall ensure the shower bases are slip-resistant and has adequate slope to direct water towards the drain. Furnish and install all new piping and associated components necessary to connect the showers to existing hot- and cold-water and waste lines. Water lines shall be PP-RCT, manufacturer to be Aquatherm or approved equal. Water lines shall connect to new thermostatic mixing (anti-scald) valves. Furnish and install wall shut off valves for the showers’ water lines.
- Furnish and install new stainless steel grab bars in the shower stalls. Components shall be installed using best practices for location placement without hindering user mobility.
- Furnish and install two (2) new floor-mounted water closets with associated plumbing fixtures and components. Water closets shall be Zurn brand or approved equal. Flush handles shall be silver in color. Furnish and install all new carriers and associated components. Furnish and install all new piping and associated components necessary to connect the water closets to existing water and waste lines. Water lines shall be PP-RCT, manufacturer to be Aquatherm or approved equal. Furnish and install wall shut off valves for the water closets’ water lines.
- Furnish and install new 6” ceramic base and 12” x 12” non-slip ceramic floor tile and associated installation materials in the bathroom extension. Furnish and install new transition strips at locations with

differing floor types. The base and tile's style, installation pattern, and color shall match existing. Grout color shall match existing. Confirm styles and colors with the Base Architect.

- Furnish and install new "6" x 6" ceramic wall tile, 4' wainscotting, and associated installation materials in the bathroom extension. The wall tile and wainscotting shall match existing. Confirm color and dimensions match with the Base Architect. Grout color shall match existing. Confirm styles and colors with the Base Architect.
- Furnish and install new shower and stall partitions, associated hardware, and mounting components. These stalls do not need to achieve ABA compliance. Color of the stall partitions shall be prefinished by the manufacturer and approved by the Base Architect.
- Furnish and install one (1) new toilet tissue dispenser in each stall. Components shall be installed using best practices for location placement.
- Any damage to the new floor and wall tiles caused by the Contractor while mounting stall partitions and bathroom and plumbing fixtures and accessories will be fixed or replaced by the Contractor, at the Contractor's expense.

2.3.27 Demolish existing light fixtures and fixture whips back to the nearest junction box. Furnish and install new 2'x2' flat panel LED light fixtures with new fixture whips. Provide all new material including, but not limited to, conduit, conductors, junction boxes, white light switches, and white dimmers to supply the new room layout with lighting. Recess mount the light fixtures in the new suspended ceiling system. Light fixtures shall comply with ASHRAE 90.1 power and lighting standards and with base, local, and national emergency lighting codes. Light fixtures shall have necessary seismic support. Place new light fixtures in locations necessary for optimal lighting qualities for essential daily functions and occupant comfort. Light fixtures shall be placed in locations that ensure minimal disruptions (i.e. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. Design, furnish, and install new white ceiling occupancy sensors or wall switch sensors and accompanying power pack relays. Components shall be compatible and connect with the new lighting circuits and light switches of each room. Furnish and install emergency exit signs and egress lighting in accordance with NFPA 101. Extend emergency exit signs and egress lighting to include the project's construction areas. Design and approval to be completed by Electrical P.E. Contractor shall test and modify the system as required. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.

2.3.28 Demolish existing lighting fixtures within both North and South Mechanical Rooms. Install new overhead LED lighting fixtures.

2.3.29 Demolish existing power outlets and receptacles at locations impacted by construction actions of this project. Test before demolition to ensure power is disconnected. Keep existing conductors. Cap, pull, coil, and tie existing conductors up overhead (above suspended ceiling system) for future use. Keep existing junction boxes, conduit, and runways in place, unless locations conflict with construction actions of this project.

2.3.30 Reuse existing electrical components to the maximum extent possible. Extend the system as required for the new room layout. Furnish all new components required for extension. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing communication and electrical systems. Furnish and install new receptacles, outlets, switches, and faceplates in existing locations. Modified rooms shall receive duplex receptacles spaced appropriately. Components shall match and be white. Components shall be compatible and reconnect with existing conductors. Contractor shall test and modify the system as required. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.
- Room 36 – In addition to the requirements of 2.3.30, install counter height duplex GFCI power outlets and receptacles appropriately spaced along the new counters. The room shall receive one (1) fully functional dedicated 240V 30A dryer circuit with GFCI protection, and one (1) GFCI washer receptacle. The dedicated dryer circuit shall use a 4-prong NEMA Type 14-30 receptacle. See concept drawings for locations.

2.3.31 Demolish existing communication locations impacted by construction actions of this project. Test before demolition to ensure power is disconnected. Keep existing conductors. Cap, pull, coil, and tie existing conductors up overhead (above suspended ceiling system) for future use. Keep existing junction boxes, conduit, and runways in place, unless locations conflict with construction actions of this project.

2.3.32 Reuse existing communication components to the maximum extent possible. Extend the system as required for the new room layouts. Furnish all new components, except for conductors, required for extension. Base Communications shall install new conductors. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing communication and electrical systems. Demolish and remove existing communication faceplates and jack. Furnish and install new faceplates and jacks in existing locations. Components shall match and be white. Components shall be compatible and reconnect with existing conductors. Contractor shall test and modify the system as required. Applies to the following rooms:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.
- Classroom 2 Specific - The new briefing room shall receive one (1) duplex telephone jack and two (2) duplex LAN jacks, at a minimum. Contractor shall ensure that the new interior partition wall that faces towards Classroom 02 will receive one (1) duplex telephone jack and one (1) duplex LAN jack, at a minimum.
- Room 5 Specific - The room shall receive two (2) duplex telephone jacks and three (3) duplex LAN jacks, at a minimum.
- Room 11 Specific - Each two-ship briefing room shall receive one (1) duplex telephone jack and two (2) duplex LAN jacks, at a minimum. Components shall match and be white.
- Room 17 Specific- Each office shall receive two (2) duplex telephone jack and two (2) duplex LAN jacks, at a minimum.
- Room 25 Specific - Each briefing room shall receive one (1) duplex telephone jack and two (2) duplex LAN jacks, at a minimum.
- Room 36 Specific - The room shall receive three (3) counter height duplex telephone jacks and three (3) counter height duplex LAN jacks, at a minimum, spaced along the new counters. See concept drawing for locations.
- Room 43 Specific - Each briefing room shall receive one (1) duplex telephone jack and two (2) duplex LAN jacks, at a minimum.
- Fitness room + 2 Adjoining Rooms Specific - The room shall receive four (4) duplex telephone jacks and eight (8) duplex LAN jacks, at a minimum.
- Step Desk Specific - The step desk shall receive two (2) duplex telephone jacks and two (2) duplex LAN jacks, at a minimum. The wall mounted counter shall receive two (2) duplex LAN jacks, at a minimum.

2.3.33 Relocate existing detectors, strobes and alarms, and mass notifications that obstruct locations of the new interior partition walls, suspended ceiling system, and electrical and HVAC components. Extend items to include the new room layout. Furnish and install items in accordance with NFPA 101 and Air Force Occupational Safety and Health Standards 91-118, reusing existing components to the maximum extent possible and reprogram fire control panels accordingly (if applicable). If these items are required in additional areas and/or existing items cannot be relocated to extend in these additional areas, furnish and install new detectors, strobes and alarms, and mass notifications to match those existing within the project's construction area. If this occurs, furnish and install conduit, wiring, associated components, and anything else required. Primary and secondary power for the smoke detectors must be provided from the fire alarm control panel. The facility's fire alarm system must always remain operational throughout this project.

2.3.34 Contractor shall coordinate communication work with the Government and Base Communication Officer (BCO) to achieve designs necessary for future implementation of a complete and functional communication system in the unit. Work shall follow UFC 3-580-01 Telecommunications Interior Infrastructure Planning and Design, and any other necessary telecommunication requirements determined by federal, state, and local regulations.

2.3.35 If any existing components of the fire alarm system require relocation or any modification, it is the responsibility of the contractor to perform whatever work is necessary. Coordinate all fire alarm work with the Government and Emergency Services to ensure proper disconnection and storage of all components. The Contractor must furnish and install all components, detectors, sensors, conductors, and anything else necessary. Any damage to fire alarm or fire suppression components caused by Contractor will be fixed or replaced at the Contractor's expense.

2.3.36 Furnish and install a single-mode fiber card to the existing Notifier 320 fire alarm panel with programming on the central.

2.3.37 Test, adjust, and balance the entire building wide HVAC system associated with air handlers 1 and 2, including rooms and zones not mentioned in the scope above. Provide a complete TAB report of the system details including but not limited to those specified in UFC 1-200-02 and UFC 3-410-01. It is the responsibility of the Contractor to obtain this independent, third-party service.

2.3.38 Demolish existing supply registers and grilles. Furnish and install all new white supply registers and return grilles. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. Components shall be aluminum with a factory finish to match the new ceiling. Applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 11, Room 17 + Corridor, Room 25, Room 29, Room 36, Room 37, Rooms 40/41, Room 43, Rooms 156/157/158/159, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjacent Rooms, Step Desk.

2.3.39 Demolish existing supply registers, return grilles, and associated flex duct, leaving in place the existing sheet metal duct trunk. Cap any take-offs not utilized in the new ductwork system for the space. Furnish and install new flex ducts and balancing dampers from the existing duct trunk take-offs to each new supply diffuser location. Extend all other remaining systems as required for the new room layout. Furnish all new components required for extension. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. All new ductwork and piping shall not be placed directly on the existing suspended ceiling system grid and tiles. All new ductwork and piping must be insulated. Ensure the new components connect to and operate the existing HVAC system to achieve occupant comfort. This applies to the following spaces:

- Classroom 1, Classroom 2, Classroom 3, Room 5, Room 17 + Corridor, Room 25, Room 37 (Mother Room Portion), Rooms 40/41, Room 165 (NCOIC Office), Room 165 (Locker Annex), Fitness Room + 2 Adjoining Rooms, Step Desk.

2.3.40 **Room 11** Specific HVAC Work:

- Demolish existing supply registers and associated flex duct, leaving in place the existing sheet metal duct trunk. Cap any take-offs not utilized in the new ductwork system for the space. Demolish return grilles.
- Furnish supply air to each briefing room as well as the newly created central hallway. Utilize door grille inserts for a return air path from each briefing room to the newly created central hallway. Furnish and install new flex ducts and balancing dampers from the existing duct trunk take-offs to each new supply diffuser location. Extend the return air ductwork to the newly created central hallway. Extend all other remaining systems as required for the new room layout. Furnish all new components required for extension. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. All new ductwork and piping shall not be placed directly on the existing suspended ceiling system grid and tiles. All new ductwork and piping must be insulated. Ensure the new components connect to and operate the existing HVAC system to achieve occupant comfort.
- Relocate existing zone temperature sensor from its existing location to a new location unobstructed by construction actions of this project. Contractor shall use best practice of location placement. Reuse existing components to the maximum extent possible, extending them to the new location. Furnish and install all new materials required for extension. Ensure the zone temperature sensor connects to and operates the system. Furnish and install all materials necessary to patch and repair existing walls as necessary at the existing and new zone temperature sensor locations.

2.3.41 **Room 24** Specific HVAC Work:

- Install additional HVAC return grilles and duct to provide an adequate return air path back to the air handling unit. Note, the wall at the north end of room 24 was an addition after that stand-alone HVAC system had been installed, blocking the return air path back to the HVAC closet.

2.3.42 **Room 36** Specific HVAC Work:

- Provide a stand-alone HVAC system for this space, independent of AHU2.
- Demolish existing supply registers and associated flex duct, leaving in place the existing sheet metal duct trunk. Demolish return grilles and ductwork. Cap all supply and return takeoffs from the main supply and return duct trunks. Demolish existing thermostats.
- Provide a heat-pump, DX, split HVAC system. Mount the indoor unit above the ceiling in room 36. Install the outdoor unit in the field outside of the Mechanical Room South’s door. Provide a concrete equipment pad for the outdoor unit.
- Install new ductwork components in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. All new ductwork and piping shall not be placed directly on the existing suspended ceiling system grid and tiles. All new ductwork and piping must be insulated.

2.3.43 **Room 37 (Men’s Room Extension)** Specific HVAC Work:

- Furnish and install a new exhaust fan, ductwork, conduit, conductors, and associated components. The new exhaust fan shall have a backdraft damper be controlled with light switch/light features, and be sized to comply with ASHRAE 90.1 HVAC standards, and with base, local, and national HVAC codes. The new exhaust fan shall be direct drive. If the motor is less than ½ HP, the fan shall be provided with an adjustable electronic speed controller.

- Demolish and cap the HVAC supply diffuser within this space. There is to be no new HVAC supply drops installed within the bathroom addition. Cooling air is intended to be pulled through the space via the exhaust fan and louvered bathroom door.

2.3.44 **Room 43** Specific HVAC Work:

- Demolish existing supply registers and associated flex duct, leaving in place the existing sheet metal duct trunk. Cap any take-offs not utilized in the new ductwork system for the space. Demolish return grilles.
- Furnish supply air to the space as well as the newly created central hallway. Utilize door grille inserts for a return air path from each briefing room to the newly created central hallway. Furnish and install new flex ducts and balancing dampers from the existing duct trunk take-offs to each new supply diffuser location. Extend the return air ductwork to the newly created central hallway. Extend all other remaining systems as required for the new room layout. Furnish all new components required for extension. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. All new ductwork and piping shall not be placed directly on the existing suspended ceiling system grid and tiles. All new ductwork and piping must be insulated. Ensure the new components connect to and operate the existing HVAC system to achieve occupant comfort.
- Relocate existing zone temperature sensor from its existing location to a new location unobstructed by construction actions of this project. Contractor shall use best practice of location placement. Reuse existing components to the maximum extent possible, extending them to the new location. Furnish and install all new materials required for extension. Ensure the zone temperature sensor connects to and operates the system. Furnish and install all materials necessary to patch and repair existing walls as necessary at the existing and new zone temperature sensor locations.

2.3.45 **Rooms 156/157/158/159** Specific HVAC Work:

- Provide a stand-alone HVAC system for this space, independent of AHU2.
- Demolish existing supply registers and associated flex duct, leaving in place the existing sheet metal duct trunk. Demolish return grilles and ductwork. Cap all supply and return takeoffs from the main supply and return duct trunks. Demolish existing thermostats.
- Provide a heat-pump, DX, split HVAC system. Mount the indoor unit above the corridor in front of room 156. Install the outdoor unit in the field outside of the Mechanical Room South’s door. Provide a concrete equipment pad for the outdoor unit.
- Install new ductwork components in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing HVAC, fire detection and alarms, and electrical systems. All new ductwork and piping shall not be placed directly on the existing suspended ceiling system grid and tiles. All new ductwork and piping must be insulated. Design duct layout to properly provide airflow with the anticipation of several rows of lockers as pictured in the concept drawings.

2.3.46 The HVAC work specified for the “Rooms 156/157/158/159” and “Room 36” is anticipated to heavily affect the function of some of the HVAC zones from AHU2. Record drawings indicate most, but not all rooms in zones 8 and 9 will now be served by independent systems. The Contractor must ensure the remaining rooms within zones 8 and 9 still receive adequate supply air and have appropriate return air paths. Supply air can come from an adjacent zone from either AHU1 or AHU2, step-desk standalone HVAC system, or a repurposed zone 8 or 9 from AHU2. Relocate thermostats where required.

2.3.47 Furnish and install a new exhaust fan dimensioned to fit in existing location for Men’s restroom north. Rebalance existing exhaust ductwork to meet exhaust rates per ASHRAE 62.1.

2.3.48 Controls Upgrades (Apply to the entire facility Building 230)

- Currently the HVAC controls communicate via N2 protocol. Convert new control sequences to Bacnet protocol. This may require additional wiring runs.

- Replace 17ea JCI VMA Controllers on each of the VAV boxes with JCI "M4-CVM" vav box controllers. Upgrade each vav controller's zone sensor to a compatible sensor.
- Replace the DX9100 Controller with a JCI "M4-CGM" Controller for the central TIMS standalone HVAC system. Replace the thermostat for this system with a newer model TEC thermostat. Contractor is to develop an updated sequence of operation.
- Replace DX9100 Controller (1ea) with JCI "M4-CGM" Controllers at AHU2. Contractor to develop updated sequence of operations.
- Replace DX9100 Controllers (2ea) with JCI "M4-CGM" Controllers at AHU1 and Boiler/Chilled water plant. Contractor to develop updated sequence of operations for each.
- Replace Variable Frequency Drives for AHU1 and AHU2.
- Replace all remaining pneumatic controls equipment with electronic.

2.3.49 Demolish and replace all chilled water piping and heating water piping within both North and South mechanical rooms. Demolition extends up to the last pipe fitting before the piping exits the mechanical room. This includes all valves, fittings, pumps, valves, actuators, insulation, etc. New pumps are to be base-mounted at ground level on new housekeeping pads. All new valves and actuators are to be electronic instead of pneumatic.

2.3.50 Demolish existing water fountains including carriers and associated components. Cap water lines to prepare for installation of one (3) new ABA compliant drinking fountain with accompanying water bottle filler. Furnish and install all three (3) new wall mounted, ABA compliant drinking fountain with accompanying water bottle filler, carriers, and associated components. Reuse existing plumbing components including rough-ins, stubs, water lines, among others to the maximum extent possible. Furnish and install all new plumbing components as required to supply the building with a complete and functional drinking fountain system. Drinking fountains shall have chilled water capabilities. Repair or replace disturbed wall surfaces within the water fountain nooks. Repaired wall surfaces shall achieve level 4 finish. Paint interior walls within water fountain nooks. Paint color(s) shall be Sherwin Williams' Accessible Beige or approved equal. Color should conform to base standards and be approved by the Base Architect. Interior walls will be painted to match the color selected by the Base Architect. At a minimum, painting should include one (1) coat of primer, two (2) coats of paint, sanding and prepping of all surfaces, and placing drop cloths on the floor.

2.3.51 The Contractor is not required to furnish and install equipment lockers for this space. The Contractor must make the spaces ready for the future installation of the lockers within Rooms 156/157/158/159, Men's Restroom North, and Room 37 (Men's Restroom Extension). Lockers will be installed by the building occupants.

2.3.52 Design and furnish a new building-wide room numbering plan. This includes all newly created and existing rooms. It includes all interior rooms within Building 230, even rooms not mentioned within the scope of work. Install new numbering plaques at the entrances to all interior rooms.

<u>CLIN 002 – CONVERT CLASSROOM 04 TO OPEN STORAGE AREA, B230 ***(Bid Option #1)***</u>

2.3.53 The three rooms in the North-east corner of the facility (Fitness Room + 2 Adjacent Rooms) that were originally intended to be converted into Classroom 04, will now be converted into an Open Storage Area (OSA) for classified information. The general work to be performed within these rooms as stated within the SOW is still to be performed, except as amended by the following scope.

2.3.54 OSA Reference Documents

- Department of the Air Force Manual 16-404, Volume 3
- Technical Specifications for Construction and Management of Sensitive Compartmented Information Facilities, Version 1.5

2.3.55 OSA Perimeter Construction Standards

- All four perimeter walls (Interior South, Interior West, Exterior North, Exterior East) are constructed of hollow CMU block with interior walls being 8” thick while exterior walls are 12” thick. Exterior walls have an additional layer of brick cladding on the outside. Interior walls currently extend to the roof deck. *8” thick hollow CMU block walls* meets the ATC sound attenuation specifications from *Technical Specifications for Construction and Management of Sensitive Compartmented Information Facilities, VERSION 1.5*. This construction also meets the wall construction minimum standards from *DEPARTMENT OF THE AIR FORCE MANUAL 16-1404, VOLUME 3*.

2.3.56 Remove all wall furring materials on the interior walls of the OSA in order to expose the CMU block walls. Repair holes, openings, or penetrations found within the CMU walls to the standards mentioned in this section. Install new wall furring materials and new drywall upon completion of modifications to the CMU wall, electrical, and communications conduit. The finished wall surface will be painted drywall.

2.3.57 Infill two of the entrance doors with CMU block, utilize the leftmost opening for the entrance door to the OSA. The entrance door shall be substantially constructed of wood or metal. For out-swing doors, hinge-side protection shall be provided by making hinge pins non-removable (e.g., spot welding) or by using hinges with interlocking leaves that prevent removal. Doors shall be equipped with a GSA-approved combination lock meeting FF-L-2740 (X-10 or approved equivalent).

2.3.58 Utility openings (HVAC duct, etc.) larger than 96 square inches (and over 6 inches in its smallest dimension) that enters or passes through an open storage area shall be hardened in accordance with Military Handbook 1013/1A (Reference (au)). In this instance, the space above the ceiling tiles but below the roof deck is considered part of the OSA. This means that utility opening hardening will occur where it passes through the CMU walls or the roof, in lieu of hardening the diffusers located in the ceiling.

2.3.59 Install a wall mounted secure lockbox cabinet to be utilized to house a SIPR-network switch and associated equipment. Cabinet is to be mounted along the north wall. Ensure proper blocking is built behind the wall prior to installing drywall. Locking cabinet must be GSA approved for storage of classified equipment, designed with rack-mounting assemblies, cooling fans, and secured cable entry points. Locking mechanism is to be via padlock and latch (padlock to be provided by others). Cabinet minimum dimensions are 2’x2’x2’.

2.3.60 Install a new dedicated electric panel to be utilized for the SIPR equipment power load. Install the panel in an accessible location along an available wall within the northern mechanical room. Run supply power from the Main Distribution Panel (MDP) using the abandoned breaker feeding “STANDBY CHILLER”. The SIPR equipment power load is approximately 5 desktop computer setups and a network switch.

2.3.61 Section 2.3.30 is to apply to this space as modified:

- Reuse existing electrical components to the maximum extent possible for the non-SIPR power receptacle loads. Extend the system as required for the new room layout. Furnish all new components required for extension. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing communication and electrical systems. Furnish and install new power receptacles and outlets in existing locations. Modified rooms shall receive duplex receptacles spaced appropriately. Components shall match and be white. Components shall be compatible and reconnect with existing conductors. Contractor shall test and modify the system as required. Place no power receptacles along the north wall. Remove and do not replace receptacles along the north wall. Repair conductors appropriately. All electrical receptacles and conductors are to be installed behind the drywall.

2.3.62 Along the north wall, provide 5-8 duplex power receptacle drops fed from the new SIPR electric panel. All conduit and boxes shall be metallic and mounted to the drywall surface. Mark, label, and color code conduit, boxes, and faceplates appropriately.

2.3.63 Section 2.3.32 is to apply to this space as modified:

- Reuse existing communication components to the maximum extent possible for non-SIPR communications components. Extend the system as required for the new room layouts. Furnish all new components, except for conductors, required for extension. Base Communications shall install new conductors. New components shall be placed in locations that ensure minimal disruptions (e.g. relocating, demolishing, performance, etc.) to new and existing communication and electrical systems. Demolish and remove existing communication faceplates and jack. Furnish and install new faceplates and jacks in existing locations. Components shall match and be white. Components shall be compatible and reconnect with existing conductors. Contractor shall test and modify the system as required. Place no non-SIPR communications components along the north wall. Non-SIPR communications components shall be installed behind drywall.

2.3.64 Along the north wall, provide 5-8 duplex communications drops fed from the secured lockbox. All conduit and boxes shall be metallic and mounted to the drywall surface. Mark, label, and color code conduit, boxes, and faceplates appropriately. The Government will make the final communications connection from the secured lockbox to the communications closet.

2.3.65 Provide a complete Intrusion Detection System (IDS) for the OSA space utilizing *Honeywell Vindicator*. The entry door shall be equipped with a vindicator access control system for entry via CAC card. The entry door shall be protected by a balanced magnetic switch that meets UL Standard 634. The OSA space shall be protected with motion detection sensors (e.g., ultrasonic and passive infrared). PCU to the Central Controller. This system must include sensors for interior motion detection, sensor for door open/close, video surveillance for interior of the room, and video surveillance for exterior of entry door. Install Premise Control Unit (PCU) within the secured lockbox of the OSA room. Provide wiring in conduit from all components of the IDS back to the PCU. Conduit is to be surface mounted on the drywall. The Government will make the final communications connection from the PCU to the communications closet.

2.4 GOVERNMENT FURNISHED EQUIPMENT/MATERIALS

The Government will not furnish any equipment or materials for this project.

2.5 SALVAGE

The Government will not salvage anything for this project.

CHAPTER 3 DESIGN REQUIREMENTS

3.1 SUMMARY

Design will consist of all site investigations and design work associated with preparing complete contract documents for this project as described in this SOW. The design documents must be developed from the functional and technical data contained in this SOW and include construction drawings, schedules, and design analysis. The Contractor will also be required to submit design documents for review as described in Paragraph 3.6 as well as final contract documents. The Contractor will be required to attend review and conference meetings as needed and requested by the Government.

3.2 NATURE/EXTENT OF DESIGN

This project requires the services of licensed Design Professionals. The project requires structural design and details for the demolition and subsequent reconstruction of several interior walls within the facility. Mechanical design and details are required for this project with split-systems, pumping, piping, valves, etc. being installed. Electrical design and details including power, conduit, circuit breakers, wiring, etc. are required. Professionally sealed drawings are required. Design/shop/coordination drawings, materials and equipment submittals, and final as-built record drawings are required for this project. Contractor must maintain accurate redline As-Builts throughout construction and must submit final As-Built drawings (with any of the redline changes updated on the drawings) upon project completion (see paragraph 4.7.2 for more information). Provide photometric analysis of all renovated lighting systems detailing compliance with all applicable codes and standards. See UFC 3-530-01 for details on the required photometric analysis. Photometric analysis is to be provided at the 35% design. Provide psychrometric/load design analysis of all renovated HVAC systems detailing the system's compliance with all applicable codes and standards. See UFC 3-401-01 and UFC 3-410-01 for details on the required HVAC design analysis. Psychrometric/load design analysis is to be provided to the Government at the 35% design.

3.3 RESPONSIBILITY

3.3.1 Contractor must perform all services, including travel, required to prepare and to furnish complete design documents consisting of drawings, analysis, and addenda as required, for the construction of the project in accordance with this SOW.

3.3.2 Record drawings (attached) are provided to the Contractor as a courtesy at the discretion of the Government, but their accuracy is not guaranteed. The Contractor is responsible for doing his own field investigations of the utilities, existing conditions, and other items in the area.

3.3.3 The Government may provide conceptual drawings to the Contractor, but their accuracy is not guaranteed. The Contractor must do his own field investigations of the utilities, existing conditions, and other items in the area and base his design accordingly.

3.3.4 Although the sizing and specification of equipment is suggested by this SOW, the Contractor is responsible for all remaining design and coordination to provide an acceptable final product that meets intent of this SOW. Contractor is responsible for ensuring constructability and code compliance.

3.3.5 The Contractor must immediately notify the contracting officer and secure any documents found with "Classified", "Controlled Unclassified Information" (CUI) or personal information. The Contracting Officer will direct disposition of these documents.

3.4 DESIGN STANDARDS

All design and construction must be accomplished in accordance with applicable UFC's, ETL's, (UFC's and ETL's may be found on-line at www.wbdg.org.) local and national codes, Columbus AFB Installation Facility Standards (IFS) and all Appendices (Special Construction, Mechanical, and Electrical) and other guidance as modified by this SOW. The IFS document may be found at the following location:

https://www.wbdg.org/airforce/ifs/columbus-afb-ifs. The list of documents includes, but is not limited to, the latest editions of the following

- UFC 1-200-01 DoD Building Code (General Building Requirements)
- UFC 1-200-02 High Performance and Sustainable Building Requirements
- UFC 3-401-01 Mechanical Engineering
- UFC 3-410-01 Heating, Ventilating, and Air Conditioning Systems
- UFC 3-410-02 Direct Digital Control for HVAC and Other Building Control Systems
- UFC 3-410-04 Industrial Ventilation
- UFC 3-501-01 Electrical Engineering
- UFC 3-520-01 Interior Electrical Systems
- UFC 3-530-01 Interior and Exterior Lighting Systems
- UFC 4-010-01 DoD Minimum Antiterrorism Standards for Buildings
- NFPA 101
- NFPA 72
- NEC 70
- ACGIH: Industrial Ventilation Manual
- Columbus AFB IFS
- CAFB Base Standards

The Contractor must submit a written request for approval of any deviations from the Government's established standards. No deviations from the Government's established standards will be permitted unless the Contracting Officer has issued prior written approval of such deviation.

3.5 DRAWINGS

Drawings under this Contract must be well prepared, complete, and accomplished in accordance with the best professional practice to show clearly and concisely the type and extent of work to be performed.

3.5.1 Drawings must be drawn to appropriate scales and dimensioned completely and accurately. Lettering must be large enough to ensure legible half-size reduction and each dimensioned drawing must have a graphic scale. Use of fractions less than 1/8-inch for building dimensions must be avoided. Use standard building material indications and symbols for architectural items and for mechanical, electrical and plumbing equipment.

3.5.2 The Contractor will use paper sheets, 24" x 36". Each sheet will be numbered consecutively beginning with "1 of (x) Sheets" and continue to the last sheet. Number individual drawing sheets as follows:

Drawing Sheet Subject	Notation	Additional Sheets
Cover (Title Sheet, Map and Index)	T1	and following
Civil/Site Work	C1	and following
Demolition	D1	and following
Landscaping	L1	and following
Architectural	A1	and following
Structural	S1	and following
Plumbing	P1	and following
Mechanical	M1	and following
Electrical	E1	and following

Note: All sheet subjects above may not be necessary for this project

3.5.3 Drawings must be prepared electronically in both AutoCAD (Release 2020) and PDF formats.

3.6 DESIGN SUBMITTALS

3.6.1 DELIVERY: Deliver two (2) hard copies of each submittal and one (1) digital copy via DoD SAFE containing each drawing sheet in PDF file format directly to the CO, along with AF Form 3000. For uploading to DoD SAFE, the Contractor must request an upload link from the CO and then upload the files using the details in the subsequent email from DoD SAFE. For the final design submittal (As-Builts in most cases), the contractor is to provide, in addition to the hard copies and digital copies via DoD SAFE, containing the PDF file as well as the associated CAD files in AutoCAD (Release 2020) .dwg file format. Hard copy submittals must be delivered to:

14th Contracting Squadron
495 Harpe Blvd, Suite 293
Columbus, MS 39710

3.6.2 SUBMITALS AT INTERMEDIATE STAGES OF DESIGN COMPLETION: To facilitate review and feedback throughout the design process, designs are submitted at various stages of completion as follows:

1. 35% DESIGN SUBMITTAL: Contractor must submit 35% design documents within 30 days after Notice to Proceed (NTP). The 35% design must include complete and thorough detailing to leave no question as to how demolition and construction is intended. All design documents must be clearly annotated (in both title block and revision block of drawings) with “35% REVIEW SET” and the date.

2. 65% DESIGN SUBMITTAL: Contractor must submit 65% design documents within 21 days after receipt of the “Approved” 35% design. The 65% design must include complete and thorough detailing to leave no question as to how demolition and construction is intended. All design documents must be clearly annotated (in both title block and revision block of drawings) with “65% REVIEW SET” and the date.

3. 95% DESIGN SUBMITTAL: Contractor must submit 95% design documents within 21 days after receipt of the “Approved” 65% design. The 95% design (essentially complete design and drawings, pending review and acceptance by Government as the 100% APPROVED CONSTRUCTION SET) must include complete and thorough detailing to leave no question as to how assembly is intended. All design documents must be clearly annotated (in both title block and revision block of drawings) with “95% REVIEW SET” and the date.

4. 100% DESIGN/APPROVED CONSTRUCTION SET: Contractor must submit the 100% DESIGN APPROVED CONSTRUCTION SET within 14 days after Contractor receipt of the “Approved” 95% design. The construction set must incorporate and/or address any Government comments on the 95% design and should leave no question to how demolition and construction will be done. If professionally stamped and sealed drawings are required for this project (paragraph 3.2), the approved construction set must include all stamped drawings. All final design documents must be clearly annotated (in both title block and revision block of drawings) with “APPROVED CONSTRUCTION SET” and the date.

5. For each holiday or extraordinary event in a submittal period, increase the required days for submittal by one day for each day in the event. Deliver no submittals within (before or after): one week of Thanksgiving; two weeks of Christmas; or one week of New Year’s Day. Extraordinary Events include: unscheduled holidays for base personnel; base-wide training exercises; etc. The Government may prohibit work during such events.

3.6.3 PARTIAL SUBMISSIONS: Partial submissions normally will not be accepted. However, under some circumstances, to expedite project execution, the Government may accept design/shop drawings or material submittals for major elements or long-lead-time equipment for review and approvals prior to 95% design submittal. Government approval of such items does not relieve the Contractor of his responsibility to ensure a complete, constructible, code-compliant design.

3.6.4 CONSTRUCTION PRIOR TO 100% DESIGN: Construction may not commence until Government approves the 100% design. Exception: To expedite contract completion, Contractor may petition CO for an Interim Notice to Proceed with specific elements of construction. Government review does not constitute approval or acceptance of any variations from this SOW, the Contractor’s Proposal, or the Contract unless authorized in writing by the CO. The Interim Notice to Proceed, if exercised, will in no way mitigate the responsibility of the Contractor to provide a complete, constructible, code-compliant design.

All drawings, specifications, design analyses and other project materials become the property of the Government.

3.7 DESIGN REVIEWS

3.7.1 DESIGN PHASE REVIEWS: Government review of each XX% design submittal (where XX = 95 or 100 as required per 3.6.2) will require up to fourteen (14) days after Government receipt. The Government will determine the sufficiency of each XX% design submittal as follows:

6. Approved as Submitted. If design is sufficient to proceed without correction to the next phase submittal, one (1) copy of AF FORM 3000 marked "Approved" will be returned to the Contractor. Contractor must advance to the next stage design submittal.

7. Approved - Except as Noted. Design submittals that have only minor errors or deficiencies will be marked in red to indicate necessary corrections. Marked materials, along with one (1) copy of AF FORM 3000 indicating "Approved - Except as Noted" will be returned to the Contractor. Contractor must revise the design documents, making the necessary corrections, and advance to the next stage design submittal.

8. Disapproved. Design submittals that are incomplete or require more than minor corrections will be marked in red to indicate necessary corrections. Marked materials, along with one (1) copy of AF FORM 3000 marked “Disapproved” will be returned to the Contractor. Contractor must revise the design documents, making the necessary corrections, and re-submit as the current stage “XX% Design – 2nd (or subsequent) Submittal”.

9. Disapproved – Major Deficiency. Examples include, but are not limited to, design submittals that fall significantly short of indicated level of completion, grossly deviate from or misinterpret the SOW, or contain errors of a nature or quantity that indicate failure of Contractor to review submittals for completeness, correctness, and coordination between trades. For such submittals, the Government will not complete a comprehensive review or provide an exhaustive list of needed corrections. Rather, one (1) copy of AF FORM 3000 marked “Disapproved – Major Deficiency” along with general reason(s) for disapproval will be returned to the contractor. The CO will issue a written warning to the Contractor that any subsequent grossly incorrect/incomplete submittals constitute evidence of inability to complete the project and may result in declaring the Contractor “In Default.” Contractor must correct deficiencies and resubmit as current stage “XX% Design – 2nd (or subsequent) Submittal”.

3.7.2 100% DESIGN REVIEW: The Government will have fourteen (14) days to verify that all review comments have been addressed and no unapproved changes have been made. Upon Government approval of the 100% design documents, the Contractor may begin construction.

3.7.3 IMPACT OF REVIEW ON PERIOD OF PERFORMANCE: Contract completion time includes time for Government review of drawings, specifications, and material submittals. Unless otherwise noted, the Government will use no more than 14 calendar days for review of each design submittal and each subsequent re-submittal of “Disapproved” submittals (paragraph 3.7.1, c or d). There will be no increase in the period of performance for review of re-submittals.

3.7.4 RESPONSE TO REVIEW COMMENTS: The Contractor is required to respond to all review comments and submit their annotated comments as a design analysis appendix in the subsequent design submittal. All comments must either be accepted and incorporated into the next design submittal or satisfactorily rebutted.

3.7.5 DESIGN ACCOUNTABILITY: The Contractor is solely responsible for accurately designing and constructing everything necessary to provide a fully functional product. Government approval of any design or design-related submittal does not waive or remove full liability from the Contractor to provide a design, facility, or final product that meets the intent of this SOW, performs as intended, and adheres to all applicable codes and standards.

3.8 CONFERENCES

3.8.1 PRECONSTRUCTION CONFERENCE: A Preconstruction Conference (Pre-Con) will be held at the Contracting Office as indicated.

3.8.2 COORDINATION/REVIEW CONFERENCES: Coordination/review conferences may be held from time to time as required. The Contractor may request such conferences when it is deemed necessary by both parties to clarify the work or expedite the preparation of plans, specifications or construction.

3.8.3 REVIEW/CONFERENCE MINUTES: The Contractor will be responsible for making memoranda of record of any conversations and minutes of any meeting with Government personnel concerning this project and forward one copy of these memoranda/minutes to each party concerned and one copy to the Contracting Officer within seven (7) days. The final version of the meeting minutes will be approved and filed at the CO’s discretion. Each copy will be numbered consecutively and chronologically.

3.8.4 MEETING LOCATIONS: Meetings will be held in person at Columbus Air Force Base. Contractor transportation costs for attending all conferences/reviews and meetings must be included in the Contractor's Proposal. There will be no teleconferences unless permitted otherwise by the Contracting Officer.

End of Design Requirements

CHAPTER 4 ADMINISTRATION OF CONSTRUCTION

4.1 PERFORMANCE CAPABILITIES

4.1.1 One week prior to the start of work under this contract, Contractor must submit in writing to the CO for review and possible approval the names and credentials of Contractor Project Manager, Contractor Site Superintendent, and Quality Control Manager. Subsequent proposed changes of approved personnel must be submitted in writing to and receive prior approval from the CO before working on the project. Each of these personnel may upon CO approval hold additional onsite duties which must not diminish their required performance. Contractor must demonstrate to the Government's satisfaction that proposed personnel exceed education, technical and experience levels of personnel described in this Contract including the following:

10. Contractor Project Manager (CPM) - Provide management of the prime contract to include subcontract purchasing and administration, review of material submittals and shop drawings. CPM must have full authority to develop cost proposals, negotiate with Government and subcontractors, sign awards and modifications, supervise project superintendents, attend weekly CO/CE/PM Contractor status meetings as well as pre-performance site visits, pre-final and final inspections. The CPM must have a minimum of 5 years previous experience as primary contract manager or equivalent experience as a primary estimator and negotiator with a contracting firm engaged in similar multi-discipline commercial construction projects. CPM must also have not less than 5 years' experience managing subcontractors.

11. Contractor Site Superintendent (CSS) - Provide on-site construction superintendent(s) to oversee all work under this contract. Site supervision must be daily (constant throughout each day) and includes submission of detailed weekly progress reports, scheduling and coordination of subcontractors and material suppliers, and attendance at all construction progress meetings, CO site visits, and pre-final and final inspections. Superintendents must have a minimum of 5 years previous experience as a project superintendent (exclusive of time employed as a tradesman or working foreman) for a general contracting firm overseeing one or more multi-discipline commercial construction projects. The superintendent must be available within 15 minutes during all normal working hours, except for such incidental errands as required by his duties. The superintendent must be responsible for the proper coordination and timeliness of the work, and for the proper workmanship of all trades; therefore, his absence from the project site without a suitable substitute Contractor representative must be considered as damaging to the Government.

12. Quality Control Manager (QCM) - Provide quality control management for the project. QCM must continually (constant throughout each day) ensure full compliance with all safety requirements, verification that work and materials in place and stored on site are in accordance with the approved construction documents, shop drawings and material submittals. QCM must submit daily status reports recording activities at each project site, prepare and coordinate material submittal sheets and shop drawing submittals, schedule and coordinate testing procedures, prepare quality control reports for construction progress and other required meetings, and attend all pre-final and final inspections. QCM must have a minimum of five years' experience in Quality Control Management of multidiscipline construction projects.

4.1.2 Contractor must have additional personnel including management, technical, service, labor or subcontractor, available as necessary to fulfill all construction contract requirements. The Government may restrict employment under this contract of any personnel identified as a potential threat to the health, safety, security, well-being, or operational mission of CAFB and its population.

4.1.3 Contractor must be available by telephone 24-hours a day, 7-days a week and upon 15 minutes' notice during normal duty hours and 30 minutes' notice during non-duty hours must meet the CO at a CO designated location.

4.2 COMMENCEMENT, PROSECUTION & COMPLETION

4.2.1 Contractor must start design and various construction phases at times specified by CO issued Notice(s) to Proceed (NTP), prosecute the work, and complete the entire work within the Period of Performance.

4.2.2 Contractor must provide a Submittal Log (Schedule of Material Submittals), Progress Schedules, Progress reports and Status Reports.

13. **Submittal Log**: See Material Submittals and Shop Drawings below.

14. **Progress Schedule:** (AF Form 3064) Submit the progress schedule to the CO within 10 days after contract award, unless otherwise specified by the CO. Subdivide total schedule into individual work items. Show completed work percentages at the end of each two-week reporting period. Progress schedule should only be resubmitted during the project duration if there's an issue with Period of Performance or if deemed necessary by the CO as described in 4.2.2.3.

15. **Progress Reports**: (AF Form 3065) Submit Project Progress reports to the CO minimum twice per month on the first working day of each month and again on the 15th. Describe the percentage of work completed during the report period. If behind schedule, report in writing what actions will be taken to regain the schedule. If the CO agrees to add days to the contract performance period, Contractor must provide an amended schedule.

16. **Status Reports**: Submit to the CO daily reports of the work status. In each report, list the tasks accomplished for that day, any deliveries received, equipment currently on-site, weather conditions for that day, and number of personnel working on each area listed by trade. Photos may be included.

4.2.3 Period of Performance: Is the period between the issuance of the NTP and the time all construction is complete, including all punch list items, and final inspection is approved by the Government. The total Period of Performance for this project is **365** days.

4.2.4 Contract Completion: At contract conclusion, vacate all work areas, including: Contractor offices, storage and staging areas, and individual work sites. Restore these areas to their condition prior to Contractor occupancy not more than 14 days from the earlier of: approved final inspection or CO notification.

4.2.5 Coordination and Project Conditions: Schedule and manage submittals, materials, products, equipment, manpower, etc. to control all parts of the work efficiently and orderly. Verify compatibility of existing building utilities with new operating equipment. Coordinate required space, supports, and mechanical and electrical work shown (even if via diagrams) on Drawings. Follow routing shown for pipes, ducts, and conduit; place runs parallel with building lines. Use space efficiently to maximize maintenance and for repair accessibility for other systems. Conceal wiring behind finish surfaces and locate fixtures and outlets with finish elements. Coordinate pre-Substantial Completion clean-up in separate building areas. After CAFB occupancy, coordinate site access to correct defective (or non-compliant with CDs) to minimize disruption of Government activities.

4.2.6 Cutting and Patching: Only employ skilled and experienced workers to cut and patch. Report in writing to the CO wherever cutting and patching might degrade: safety; appearance, structure, weatherproofing, O&M, or separate Contractor or government construction. Prevent damage to existing or new construction and provide proper surfaces to receive patching and finishing. Match new finishes to existing adjacent surfaces unless noted otherwise. Maintain integrity of existing walls, ceilings, or floors; completely seal voids. Refinish entire assemblies and continuous surfaces to nearest intersection. Cut, fit, and patch where CDs require e.g.: mechanical and electrical penetrations; integration of new components with one another or with existing construction; removal and replacement of defective and non-conforming elements; remove samples for testing.

4.2.7 Special Procedures: Employ skilled and experienced tradespersons to perform alteration work. Remove debris from open work areas and concealed spaces. Remove, cut, and patch to minimize damage and provide means of restoring products and finishes to original or specified condition. If a change of plane of 1/4 inch or more is discovered, submit recommendation to the CO for providing a smooth transition. Patch or replace portions of existing surfaces which are damaged or showing other imperfections. Finish surfaces as specified.

4.2.8 Airfield Waiver: Airfield waiver is not required for this project.

4.2.9 Welding Permits: No cutting or welding will be permitted without first obtaining a permit issued by the Base Fire Department. All fire and safety regulations are to be followed stringently. Contractor's failure to obtain a cutting or welding permit may not be reason for extending the contract performance period. Any damages incurred while welding work is done without a permit are the responsibility of Contractor.

4.3 SCHEDULING

4.3.1 Contractor must coordinate all work schedules and manage progress of work with the CO and CEPM prior to start of work. Weekly progress meetings will be conducted with the CO, CMI, and Contractor's team which may require corporate management representation at CAFB discretion.

4.3.2 Contractor must schedule all work. Performance periods must not be established or extended to accommodate insufficient personnel resources. Contractor will not move crews on and off the project while making minimal progress on concurrent projects as this must be construed as having an inadequate work force to fulfill the contract requirements. Contractor's CPM and CSS must not change more than two (2) times during project construction as this would demonstrate an inadequate management force to fulfill contract requirements.

4.3.3 Before construction begins, Contractor must agree with CO on a sequence of procedures; means of access to premises and buildings; materials and equipment storage space; delivery of materials and use of approaches, corridors, and stairways. Contractor may be required to revise the CD work-phasing schedule.

4.3.4 Interference and inconvenience to Government operations and personnel caused by project work including use and delivery of materials, tools and equipment must be kept to a minimum.

4.3.5 The Government may require that work, so far as practicable, be done in separate phases. Each phase may relate to a different occupied or unoccupied area, in which work in one area must be completed before work in another area may begin. Phased work areas must be clearly delineated and annotated in the CDs.

4.3.6 Hours of Work: Normal work hours will be Monday through Friday from 7:30am to 4:15pm excepting days observed as Federal Holidays as listed below:

New Year's Day	1 January
Martin Luther King's Birthday	Third Monday in January
President's Day	Third Monday in February
Memorial Day	Last Monday in May
Juneteenth Day	19 June
Independence Day	4 July
Labor Day	First Monday in September
Columbus Day	Second Monday in October
Veterans Day	11 November
Thanksgiving Day	Fourth Thursday in November
Christmas Day	25 December

1) Observe Saturday holidays the preceding Friday. Observe Sunday holidays the succeeding Monday.

2) For weekend, after hours, or holiday work, submit a 72 hour advance written request to the CO, describing dates, locations, and types of work. CAFB has the right to approve or deny the request. Work in CAFB occupied areas during extra-normal work hours require AF escorts in the occupied areas. Modification to contract sum or period of performance will not be allowed.

3) The Government's request for Contractor to work outside of the normal duty hours must be justification for modification to contract sum or period of performance.

4) If CAFB observes an unscheduled holiday, participates in training exercises, etc. Contractor may be prohibited from working for part or the whole of a day. 14CES does not know how many or when these activities might be scheduled but anticipate not less than two per year. CMI will allow additional contract time for these disruptions. Zero-cost modification to period of performance will be allowed.

4.3.7 Construction Schedule: Submit preliminary schedule to the CO at the Preconstruction Meeting, but not later than fourteen (14) days after issuance of Notice to Proceed. After CAFB review and comment, revise and resubmit schedule within seven (7) days. Upon approval by the CO, the schedule becomes the official Construction Progress Schedule. Keep a copy available at Contractor (field) Office. This schedule will only be changed with prior approval of the CO through a contract Modification.

4.3.8 Construction Progress Schedule: Submit and maintain an accurate and current Gantt chart to track construction progress with a separate horizontal line for each major work activity of subcontractors, manufacturers, and suppliers. Vertical lines must identify the first workday of each week. Show complete sequence of construction by activity, identifying separate stages of work and other logically grouped activities. Indicate early and late start, early and late finish, float dates, and duration of each activity. Revise chart weekly to accurately identify progress made, percentage completion, and projected completion date for each activity. Show activities modified and scope changes. Submit written report to identify problems, anticipated delays, corrective actions, and affected completion dates. Provide additional progress schedule information as requested by the CMI. Construction progress schedule update forms will be provided at the preconstruction conference.

4.4 MANAGING

4.4.1 Preconstruction Conference: at CAFB will be scheduled by the CO after issuing the NTP. Required attendees include the Contractor, Contractor Project Manager (CPM), and Contractor Site Superintendent (CSS). Contractor must record meeting minutes and within two days after the meeting distribute two (2) copies to all participants and to all others affected by decisions made at the meeting. The meeting Agenda will be issued in advance by the CO and will include one or more of the following:

17. Execution of Government-Contractor Agreement.
18. Submission of executed bonds and insurance certificates.
19. Distribution of Contract Documents.
20. Submission of Subcontractors list, products list, schedule of values, and progress schedule.
21. Designation of personnel representing the parties in Contract.
22. Procedures for processing field orders, submittals, substitutions, payment applications, proposal requests, Change Orders, and Contract closeout.
23. Construction Schedule

4.4.2 Site Mobilization Meeting: Prior to Contractor project site(s) occupancy, the CO will schedule a meeting at which required attendants include the CO, Contractor, Contractor's Project Manager (CPM), Contractor's Site Superintendent (CSS), and major Subcontractors. Contractor must record meeting minutes and within two days after the meeting distribute two (2) copies to all participants, and to all others affected by decisions made at the meeting. The Agenda for this meeting will include:

24. Government and Contractor Use of Premises.
25. Government Requirements.
26. Construction facilities and controls provided by Government.
27. Temporary utilities provided by Government.
28. Security and Housekeeping Procedures.
29. Construction Schedules.
30. Application for Payment Procedures.
31. Testing Procedures.
32. Record Documents Maintenance Procedures.

4.4.3 Weekly Progress Meetings: with the CMI will occur throughout construction duration. Required attendants include the CO, Contractor, the CPM, and the CSS. Contractor must record meeting minutes and within two days after the meeting distribute two (2) copies to all participants, and to all others affected by decisions made at the meeting. The agenda for these meetings will include the following:

1) Review previous week's work progress.	Review minutes of previous meetings.
2) Evaluate work quality and safety.	Review field observations, problems, and decisions.
3) Construction Schedule update.	Corrective measures planned to regain projected schedules.
4) Evaluate construction progress.	Identification of problems which impede planned progress.
5) Work progress planned for the next week.	Effect of proposed changes on construction schedule
6) Submittals: review status and schedule.	Review of off-site fabrication and delivery schedules.
7) Other business	

4.5 MATERIAL SUBMITTALS AND SHOP DRAWINGS

4.5.1 Definitions: A **submittal** is a package of project information, samples, drawings, schedules, certifications, product data, etc., submitted to the Contracting Officer for Government review. A **deviation** is a submittal wherein Contractor identifies an item that is in agreement with the intent of the Contract Documents, but does not precisely conform to those documents and Contractor requests either substitution or change for the purpose of increasing the quality of the final product. A Submittal or Deviation of a system must be considered as an integrated collection of component parts.

4.5.2 Purpose: Submittals formalize Government review of Contractor choices in complying with the CDs and providing Minimum Installer Qualifications (Section 4.8.1). They also formalize Contractor's proposed Deviations, color choices, shop drawings, etc. early enough in the contract time when changes will have less impact on the ordering of materials and products.

4.5.3 Accountability: Contractor is wholly responsible for the contents of a submittal. Contractor by act of submittal certify that all items listed or implied, fully meet the intent, materiality and requirements of the design, quality, and functionality of the CDs. Neither Contractor furnishing nor Government approval of a submittal either suspends or waives Contractor responsibility for full compliance with the CDs. This includes materials, equipment, equipment sizing, performance, etc. The Contractor maintains sole responsibility for providing a fully functional product that meets the intent of this SOW, performs as intended, and adheres to all applicable codes and standards.

4.5.4 Basic Compliance: Comply with all "Material Submittals and Shop Drawings" requirements of the SOW. All submittals must be delivered to the Contracting Officer within twenty-one (21) days of approval of the 100% design unless otherwise specified by the CO. Failure to deliver all submittals within this time period constitutes basis for the CO to refuse Progress Payments until all submittals are delivered to the CO.

4.5.5 Submittals must be accurate, easily legible, with all detail necessary for a thorough review. Every copy of Product data must clearly identify proposed models, options, and relevant design data including physical, functional, and utility connection requirements. Where practicable, render manufacturer data specific to this Project.

4.5.6 Contractor must review, approval stamp and sign **every** submittal before delivery to the CO. Each submittal package and shop drawing sheet must bear appropriate Design Professional(s) stamps and approval signatures before delivery to the CO. Each submittal must be attached to a Material Approval Submittal (MAS) Form (AF Form 3000). Upon Government review, one copy of each submittal will be returned with its MAS form marked to indicate approval or disapproval.

4.5.7 Government Review: Contractor must deliver two (2) copies of each submittal. Contractor must allow fourteen (14) days for government review excluding delivery time from and to Contractor.

4.5.8 Submittal Log: Contractor must during the design phase of the project log all construction submittals required by completing a "Schedule of Material Submittals" (AF Form 66). The form must include all submittals necessary to insure the project is built to the satisfaction of CAFB including shop drawings, manufacturer's literature, product data, certificates of compliance, material samples, finish samples, extensions to the design, guarantees, test results, etc. The CO will provide Contractor with an AF form 66 template. Contractor must complete and submit the AF Form 66 in MS-Excel electronic format and in hard-copy format to the CO for review and approval by the CEPM within fourteen (14) days upon issuance of the NTP. Form 66 will be the first submittal item and must be attached to its own (MAS) form. Place the entire Submittal Log in an appendix of the final specifications.

4.5.9 Proposed Products and Qualified Installers List: Within 28 days after Notice to Proceed is issued, submit list of major products proposed for use, with manufacturer name, trade name, model or catalog number designation, and reference standards (for products specified only by reference standards) of each product.

4.5.10 Material Approval Submittal: The CO will provide Contractor with MAS form template. Each MAS form must be attached to one and only one submittal item. Multiple submittal items on a single MAS form will be rejected without review. The MAS "Submission Number" must be the same as its corresponding "Item No." in the Submittal Log. Each submittal must reference the specifications paragraph or the drawing sheet number where the submittal is required. Clearly mark an MAS form "DEVIATION" for any whole system or any part thereof to which the Deviation applies. The Deviation must be reviewed by the CEPM.

4.5.11 <u>Re-Submittals</u>: Resubmitted MAS forms must bear the original "submission number" with an alphabetic suffix (sequenced for multiple re-submittals). Identify all changes made since the previous submission. Allow fourteen (14) days for Government review excluding delivery time from and to Contractor.

4.5.12 <u>Materials, Finishes, Patterns, Textures, Colors, etc.</u>: Submit samples of all visible exterior or interior finish items including: surfaces, materials, paint, fixtures, doors and frames, windows, glass, and hardware prior to ordering or installation. Contractor must deliver the full range of available manufacturer samples to the CEPM showing all materials, finishes, patterns, textures and colors available for each item. Submit ALL samples at the same time. Attach to each sample an AF Form 3000 with Opportunity ID Number, Project Name and Submittal Number. The CEPM will make selections and will keep one sample of each item submitted. Any non-compliance with this paragraph voids any selection regardless of CEPM approval to the contrary. Samples must be the actual material, or actual material coatings bonded to the actual backup material that will be installed. Submittals with false or approximate material or color renderings may be rejected. Sample sizes must be sufficient for the CAFB to perceive true colors and patterns. If a submitted material does not meet the design objectives, CAFB reserves the right to reject the product and require a new submittal. The CEPM decision in this regard will be final.

4.5.13 <u>Certifications</u>: Submit all Contractor, subcontractor, fabricator, and manufacturer certifications for review and, if acceptable, approval by the Government. Denote the specified requirements met or exceeded by the certified element provided whether material, construction system, portion of work, product, equipment, etc. Include supporting references, data, affidavits, certificates, etc. Certifications may be recent or previous test results on the same provided element, but must be acceptable to Government.

4.5.14 <u>Manufacturers Information</u>: For each manufactured product provided in this Project, submit three copies of all manufacturers' printed instructions for delivery, storage, assembly, installation, adjusting, finishing, and usage to the Government. Submit three copies to the CO of any manufacturer's field report within 30 days of manufacturer's field visit and observations.

4.5.15 <u>DD Form 1354</u>: Contractor must submit a completed DD Form 1354, *Transfer and Acceptance of Military Real Property* to the CO prior to final punch-walk.

4.6 WORK BY GOVERNMENT

4.6.1 The Government reserves the right to accomplish work using Government or Contractor work forces other than those contracted for the Project, as the Government deems necessary or desirable, and so doing will not breach or otherwise violate the Project contract. The Contractor may be required to coordinate work with one or more other Contractors in order to complete the Project.

4.6.2 Contractor must bear responsibility for any existing Government material or equipment to be removed and reinstalled. Contractor must protect against damage or loss any Government equipment remaining at construction sites. Government equipment damaged or lost while stored or moved by Contractor must be replaced by Contractor with equal or better equipment.

4.7 PRE-FINAL AND FINAL INSPECTION

4.7.1 The Contractor may request in writing a pre-final inspection purposed to obtain Government assistance in identifying potential problems prior to final inspection. However, any discrepancies (punch list items) identified at the pre-final inspection shall be completed before the final inspection. The Contractor shall schedule the pre-final inspection at least ten (10) days before the final inspection.

4.7.2 If the Contractor does not request a pre-final inspection, the Contractor must schedule final inspection NOT LATER than 10 calendar days prior to the contract completion date unless precluded by Government scheduling problems. Before requesting final inspection, Contractor must submit to the CO project close-out documents including: hard-copy and digital copies (delivered via DoD SAFE) as-built drawings, all test and recycling reports, O&M manuals, DD1354 package, and a quality control (QC) report signed by his Quality Control Manager (QCM) listing any discrepancies. Contractor must submit a written request for final inspection to the CO a minimum of 36 hours prior to the requested inspection date. The request must include certification that Work is complete according to the CDs and ready for CO review. If the CO approves the request, Submit final Payment Application identifying total adjusted Contract Sum, previous payments and payment remaining due.

4.7.3 The QCM, Contractor's superintendent, and CO will conduct the final inspection. Contractor and subcontractors must correct discrepancies and punch-list items within the time limit specified by the CO. In undertaking a final inspection, if the Government determines the project unsuitable for final inspection, the CO will end the inspection, and not less than 24-hours must pass before the final is rescheduled. If, due to one or more such terminations, the actual final inspection date is later than the contract completion date, Liquidated Damages (LDs) may be assessed before the rescheduled "final" inspection. Contractor must submit each request for rescheduled final inspection in writing to the CO.

4.7.4 The CMI and CEPM are the designated representatives of the CO for the purpose of technical surveillance of workmanship and inspection of materials for work performed under this contract. This designation in no way authorizes anyone other than the CO to obligate the government to changes in the terms of the contract. All field changes must be approved by the CO prior to accomplishment. Government inspections or related comments neither constitute QC nor in any way either substitute or supplement Contractor's QCM responsibility. CMI will conduct inspections, but only the CO may authorize final acceptance.

4.8 WARRANTY

4.8.1 Minimum Installer Qualifications: Prerequisite to all warranty and required on every aspect of this project are demonstrated Minimum Installer Qualifications including: manufacturer certification (if required by manufacturer's warranty) that the installer is qualified to perform the work specified.

4.8.2 Standard Warranty: Provide a materials and workmanship warranty for all work for not less than one year from time of total project acceptance by the government.

4.8.3 Exceptions to Standard Warranty: Where this SOW requires longer warranties for one or more products or systems, the extended warranties will supersede the Standard Warranty, but in no case will any item in the project be warranted for less than the Standard Warranty. Whenever products or systems offer a warranty greater than the Standard Warranty, Contractor must pass the greater warranty in full to the Government.

4.8.4 Equipment Data:

1) Major Equipment: Provide a list of all equipment furnished and installed under this contract. This list must include each piece of equipment having a serial number. Each listing must positively identify the piece of property by including all the following information as applicable: date installed/replaced, warranty/guarantee expiration date, item installed, type, model, serial number, style, voltage, cycles, horsepower, size, quantity, frame, item cost, item replacement cost, and location of item/equipment. This list must be furnished to the CO as one (1) reproducible and three (3) copies at the CO's request any time during the contract

2) Major equipment includes air conditioners, air handlers, transformers, and electric motors, compressors, condensing units, chillers, exhaust fans, generators and transfer switches. Contractor

must place an Equipment Warranty sticker on all equipment furnished and installed under this contract.

3) This is not meant to include: light switches, fixtures, relays, valves, and such material items as: piping, insulation, and minor component parts of larger assemblies.

4.8.5 Contractor and Subcontractors must perform any warranty related work according to all manufacturer specifications and recommendations so as not to reduce or void any warranty. Contractor must transfer all manufacturer warranties to the Government on a submittal AF Form 3000 prior to project closeout.

4.8.6 Emergency Repair: Failure of any mission essential work under warranty constitutes an emergency, and Contractor must complete corrective repair(s) not later than a date to be established by the CO. If not responsive in a timely manner, Contractor may be charged for Government to complete repairs.

4.8.7 Non-Emergency Repair: Contractor must respond within 48-hours and effect corrective action in such timely manner as to minimize down-time and mediate inconvenience to any CAFB employees. CO's determination of appropriate time to complete repair(s) governs.

4.8.8 Contractor Non-Responsiveness: If Contractor fails to respond to notifications, the CO will find repair methods and seek restitution through legal means, including through Contractor bonding agents.

4.9 CONTRACTOR OFFICE

4.9.1 No CAFB building will be available for Contractor who requires a temporary office on CAFB. At the Pre-Construction Conference, the CO will designate a site for Contractor's installation of a temporary office on CAFB (not less than 30' from existing structures). Proposed improvements including extensions of utility lines into this area must be approved by the CO prior to installation. Contractor must bear all expenses of these improvements and temporary office. Prior to contract completion, remove buildings, foundations, and utility services and restore all areas thereby affected to their original condition.

4.9.2 Contractor's office must be a weather tight standard manufactured structure securely fixed to foundations, with steps and landings at entrance doors, having a clean appearance and color that are aesthetically acceptable to the CEPM with no patches, broken windows, dirt, rust, etc. with trailers having skirts to grade. Contractor must bear the responsibility and expense for all site work and setup.

4.9.3 Provide and maintain the office with services and equipment including working HVAC equipment to automatically maintain 68º heating and 76º cooling, electric lighting, electrical receptacles, fire extinguishers, and adequate lighting (50 fc at desk top height).

4.9.4 Provide office with items including a facsimile machine, Email address, and telephone service (may be cell) so Contractor employees may be contacted at all times during normal duty hours. Provide CO with an emergency phone number so Contractor can be reached at all other times. All installation, connection and operation costs of these items including utilities, phone, computer, and internet and office security must be provided at Contractor's expense. All on-base connections may be coordinated and monitored by the CO.

4.9.5 Contractor must furnish its own housekeeping and janitorial services for office space. At all times maintain in clean and orderly appearance the interior and exterior of Contractor base office. Trees and shrubs must be kept trimmed and regular mowing must keep grass and other ground cover to not more than 4" height.

4.9.6 Contractor must provide a commercially produced white painted metal job sign on the office bearing Contractor's name and project name in four-inch high black lettering. Job sign shall include up to date information, including current construction completion date.

4.10 CONSTRUCTION SITES

4.10.1 Temporary Power: Provide all necessary electrical connections including temporary transformers, utility poles, cable, weather-heads, panels, and any other electrical items needed for construction. Final electrical hookup must be done after approval of the Contracting Officer, and with a minimum written twenty-one (21)-day notification of hookup. All electrical work must conform to the latest editions of the National Electrical Code (NEC) and the National Electrical Safety Code (NESC). Any hookups to the exterior electrical system (anything outside building demarcations) will be done by or under the inspection 4-County. Requests for any exterior work must be submitted to the 14CES Electrical COR not less than 21-days in advance. Holidays are not included in the 21-days.

4.10.2 Neat and Orderly: Maintain construction sites neat, clean, and orderly. At each day’s end collect job site debris, waste, and rubbish, and dispose of properly and remove tools and equipment from passageways. Do not leave or enclose debris, waste or rubbish in pipe chases, plenums, attics, or other closed or remote spaces. Contain dust within project site (plastic barriers) and prevent it from moving into occupied areas. Broom and vacuum interior areas prior to start of surface finishing, and keep dust away from finishes. Contractor must bear final clean-up costs. At end of project, return all construction-affected non-project elements to original condition (including accesses, grass, dirt, sprinkler systems, etc.).

4.10.3 Prevent pests and insects from entering the building renovation areas. Keep all facilities, equipment, and vehicles fully serviced and usable. Immediately remove from site rusted, broken, torn, bent, or otherwise objectionable elements, equipment, material, dumpsters or vehicles. Only normal operator maintenance on Contractor vehicles is allowed on site. Maintain trees and shrubs trimmed, grass regularly mowed and other ground cover not exceeding 4” in height. Clippings or mowing debris must not at any time be allowed to accumulate. If the CO determines the site is unsafe due to clutter or debris, the CO may immediately halt construction and the site must be cleaned by Contractor without delay to the project deadline.

4.10.4 Parking: As approved by the CMI, locate temporary Contractor parking and restrict construction parking to these areas only. Do not allow heavy vehicles, construction equipment or staging in parking areas. Keep parking areas free from mud, dirt, snow, ice, etc. Repair damaged paving including air breaks, potholes, low areas, standing water and other deficiencies, to maintain paving and drainage in their original condition. Repair all adjacent areas damaged by use of parking, to their original condition.

4.10.5 Noise Control: Contractor must comply with all applicable state and local laws, ordinances, and regulations relative to noise control. The Government may require that operations that generate excessive noise be scheduled at other than standard work hours.

4.10.6 Site Storage: As approved by the CMI, locate an exterior area to store all equipment and supplies in a manner that precludes theft or damage of any kind (including mechanical and climatic). Stored materials must be sorted, separated and neatly stacked to protect materials from wind, rain, and contact with ground. Provide adequate lighting for inspection of materials and products. Equipment and vehicles must be located in one area of the site. Contractor must provide a new six-foot high chain-link security fence (and an optional top outrigger of three-strand barbed wire) with vehicular and pedestrian gate(s) with locks. Keep vegetation clear from the fence at all times.

4.10.7 Dumpsters: Trash, debris or other refuse within the work areas must not be visible, but must be fully placed into (never overflowing) approved dumpsters or other manufactured disposal receptacles. Place receptacles within work areas, inconspicuous from the main roads, as approved by the CMI. Refuse from the project must not be dumped into government dumpsters. Construction refuse found in government dumpsters must immediately be removed either by Contractor or by the CAFB at Contractor’s expense as directed by the CO. Before the end of project and prior to final inspection, the receptacles must be removed by Contractor who must restore exposed dumpster-covered surfaces to equal or exceed their original condition.

4.10.8 Hauling Trash and Debris: Contractor must dispose of all trash and debris via sanitary landfill or other approved method conforming to all local, state, and federal guidelines and regulations. All refuse removal trucks must be covered and secured prior to leaving job sites. If any materials fall from the trucks, they must be stopped and reloaded to prevent falling debris. Contractor must bear all liability for any damage or injury resulting from falling debris. Certified dump tickets, including cost, for all waste disposals marked with the project number must be submitted to the CO for all waste disposals.

4.10.9 Access and Haul Routes: Propose and obtain prior CO approval for hauling and site access thoroughfares and confine construction traffic to these routes. Provide traffic control at CO designated areas of haul routes to minimize interference with normal CAFB traffic. Off-site streets and parking lots soiled by mud, dirt, debris, foreign objects, or spills caused by Contractor must be cleaned off the same day. Keep parking aprons used as access to work or staging areas clean and clear of all foreign objects and debris at all times.

4.10.10 Vehicular Access: Build any necessary temporary all-weather access roads from existing pavement to serve construction areas, of width and load bearing capacity to provide unimpeded construction traffic with prior approval of the CO. Provide and maintain 20 foot wide driveways with unimpeded access to fire hydrants, control valves and for emergency vehicles. Remove mud from vehicles before they enter paved areas.

4.10.11 Concrete Truck Cleaning: CAFB has no approved concrete truck clean-out sites. Washing-out of on-site concrete trucks may be done only into Contractor-owned dumpsters. If concrete trucks are cleaned out anywhere else on base, clean-up charges will be billed back to Contractor.

4.10.12 Trees, Shrubs, and Hedges and other Plants: Provide barriers to prevent injury to any plant not designated for removal within or near the project site or haul routes. Do not fasten ropes or cables to any existing trees. Vehicles or equipment must never be driven or parked within the drip line of any tree. Obtain written approval from the CO before removing or pruning any plant. Smaller plants including shrubs, hedges or trees damaged during construction, must be replaced with plants of equal size, type, and value with a one-year warranty at no expense to the Government. Larger trees damaged during construction must be recompensed with a number of smaller trees with an aggregate area of noon-shade equal to the larger tree. Damaged or destroyed plants must be replaced between 1 November and 1 April.

4.10.13 Special Excavation Requirements: Contractor must call Mississippi 811 prior to any excavation and must provide drawings showing exact location and description of the proposed excavation. The Government will help the CPM/QCM obtain excavation permits. The government will make a reasonable effort to locate and identify buried lines. "In the area" means "three (3) feet in all directions", except when the CO indicates otherwise. Any lines flagged or painted on-site or identified on the drawings or digging permit must be avoided by Contractor whenever possible.

Extra care must be taken when work is near marked underground lines and, if these lines are broken Contractor must bear all excavation or repair expenses. If a line is broken that was not by the Government either marked on site or identified on drawings or permit, repair costs will be borne by the Government, unless the CO determines otherwise. If Contractor is deemed responsible for the damaged line, Contractor must **IMMEDIATELY** repair the line to its prior condition at no cost to the Government. Contractor must maintain all marks provided by Base CE for the duration of the Project. Any utilities discovered during excavations that are not shown on the drawings must be drawn and annotated on the As-Built drawings.

Contractor must immediately notify the CO when any utility line is damaged. Contractor must immediately notify the fire department in the event that any gas line is broken or damaged.

4.10.14 Occupancy: Building 230 will not be occupied by personnel during construction. Temporary utilities (power, HVAC, lighting, etc.) will not be required to support occupancy functions during construction.

4.11 CONSTRUCTION EQUIPMENT

4.11.1 All Contractor-supplied equipment is the sole responsibility of Contractor. The Government is not liable for theft, vandalism, or damage to any Contractor supplies or equipment on CAFB. Secure and protect all materials and equipment from damage. Do not leave equipment unattended anywhere on CAFB.

4.11.2 Equipment and Vehicles: Those used on base must be safe and in good operating condition. The CMI may at any time inspect and reject any on-base equipment they consider unsafe, in poor operating condition, or inappropriate for work. Immediately notify the CO of broken down Contractor equipment on CAFB. Move broken down equipment to a CO-designated location within 24-hours. Keep any construction equipment with metal wheels or tracks (i.e. roller, excavator etc.) off CAFB paved areas. All equipment must be trailer hauled to and from construction sites. Prevent loading pavement beyond design capacity which is greater than or equal to the legal capacity of local roads accessing CAFB.

4.12 AVAILABILITY OF UTILITY SERVICES AT JOB SITES

4.12.1 Water: Subject to available supply, the Government, without charge to Contractor will from existing outlets and supplies furnish reasonable amounts of potable water. Conserve water. Provide temporary pipe insulation to prevent freezing. Contractor, at its own expense, must install and maintain necessary temporary connections and distribution lines and must remove the connections and lines prior to final acceptance of construction. Water services may not be available at or adjacent to Contractor's staging, storage or office areas, but may be obtained from an outlet as designated by the Contracting Officer.

4.12.2 Electricity: Exterior supply is available subject to approval from 14CES Electrical COR. Request hookups to exterior electricity not less than 21-days in advance. Costs will be negotiated with the CO. Electricity from interior outlets and supplies are available without cost. Do not disrupt CAFB service without prior CO approval. With CO approval, Contractor must, at its own expense, install and maintain temporary connections and distribution lines and must remove the connections and lines prior to final acceptance of construction. Do not use permanent convenience receptacles during construction.

4.12.3 Sanitary Provisions: Contractor must, at its expense, provide and maintain temporary facilities and necessary appurtenances and must remove same prior to final acceptance.

4.13 TEMPORARY UTILITIES

4.13.1 Temporary Electric Wiring: must meet the requirements as established below and must be installed, maintained, and removed by Contractor at no expense to the Government. Skilled electrical tradesman must accomplish work. Do not disrupt any CAFB use of service without prior CO approval.

4.13.2 Temporary Power and Lighting: Contractor must provide construction power according to the safety requirements of the National Electric Code, NFPA 70. Enforce all electrical safety requirements for subcontractor work. All 15 and 20-Amp outlets not part of the permanent building or structure wiring, must have ground fault circuit interrupters (GFI) for personnel protection. GFI must be provided for extension cords and for all permanent receptacles that are not properly grounded. Provide and maintain construction lighting of not less than 2 watts/ft^2, exterior staging and storage lighting of not less than 1 fc, and after-dark interior lighting of not less than 0.25 fc. Do not use permanent building lighting during construction.

4.13.3 Heating and Cooling: Provide and maintain heating and cooling devices needed to maintain construction operations. Provide these devices with regular preventative maintenance including new filters, lubrication, and parts replacement. Maintain a maximum ambient temperature of 80ºF and a minimum ambient temperature of 55ºF where construction is in progress.

4.13.4 Construction Tools and Equipment: Superseding other requirements, temporary wiring conductors installed for operation of construction tools and equipment must be either Type TW or THW contained in metal raceways or must be hard usage or extra hard usage multi-conductor cord. Temporary wiring must be secured in a workmanlike manner above the ground or floor without obstructing movement of personnel or equipment. Open wiring may only be used outside of building and according to the provisions of the National Electrical Code.

4.13.5 Removal Prior to Substantial Completion: Remove temporary utilities, equipment, facilities, and materials, prior to final inspection. Remove underground installations to a minimum depth of 2 feet. Grade site as required. Clean and repair damage caused by installation or use of temporary work. Restore to original condition existing and permanent facilities used during construction.

4.14 CONSTRUCTION SECURITY

4.14.1 Security Program: Protect Work, including construction site, construction office and all construction material, tools and equipment, from damage, theft, vandalism, and unauthorized entry until project is turned over complete to the Government. Submit a construction security program coordinated with CAFB's existing security services, and obtain CO approval for project mobilization. Maintain program throughout construction until CAFB occupancy.

4.14.2 Entry Control: Restrict entry of persons and vehicles into Project site(s) to authorized persons with proper identification. Coordinate access control with CAFB security forces.

4.14.3 Barriers: Provide barriers to prevent unauthorized entry to construction areas and to protect existing facilities and adjacent properties from damage from construction operations and demolition.

4.14.4 Use of Electromagnetic Emission Devices: Electromagnetic emission devices include: radio/RADAR transmitters, navigational aids, instrumentation's, signaling, intrusion detection and identification devices, mobile and fixed business radio communication equipment, and Military Affiliate Radio Station (MARS), Citizen Band (CB) and amateur radio stations. Contractor must comply with the AFMTC Regulation Sup 1 to AFR 700-14 which is available for inspection at the Network Control Center, Bldg. 932, CAFB, MS.

4.14.5 Historical and Archeological Finds: Contractor must carefully preserve all items having any apparent historical or archeological interest, which are discovered in the course of any construction activities. Contractor must cordon off the discovered items from further disturbance and must immediately report the find to the CO so that the proper authorities may be notified. Any items of value or interest will be considered property of CAFB.

4.14.6 Airfield Requirements:

Contractor must contact 14 Civil Engineering Squadron, Columbus AFB, or Base Operations for construction restrictions involving the flight line, taxiway and runway areas and must comply with AFI 13-213.

Two-way radio contact on VHF Radio is required for all vehicles, equipment and personnel working on the flight line, taxiways and runways. The Government will provide radios when they are required.

4.14.7 Contractor Restrictions:

33. Unauthorized Reconnaissance: Contractor access to areas outside of the immediate work area (excluding restrooms near the work site, public eating facilities, direct haul and access routes, CO and CMI, and points of supply and storage) is prohibited. Persons engaged in unauthorized reconnaissance of other Contractor or government activity will be referred to the CO for disposition.

34. Cameras: Cameras and photographing are forbidden on CAFB except by prior written approval of the CO.

35. Contractor Parking: Contractor and Subcontractor vehicles must not be parked on grass surfaces at any time. In addition, vehicles may not be driven over grass surfaces, unless indicated during site visit or pre construction conference or specifically approved by the CO.

36. Elevators: Contractor may not use elevators for any reason without prior approval of the CO.

37. Toilet Facilities: Contractor personnel will not use building facilities.

38. Smoking in AETC Facilities: The Base Commander has placed restrictions on smoking of tobacco products in AETC facilities. All Contractor and Subcontractor employees and visitors are subject to the same restrictions as government personnel. Smoking is permitted only in designated smoking areas. AFI 40-102, Tobacco Use in the Air Force, and its AETC supplement 1, outline the procedures used by the Commander to control smoking in AETC facilities.

End of Administration of Construction

CHAPTER 5 QUALITY REQUIREMENTS

5.1 QUALITY CONTROL PLAN

5.1.1 Present to the CO a Quality Control Plan (QCP) that ensures all elements of Work meet required standards and adhere to the CDs. The plan must include: who is responsible for quality inspections, processing acceptance or rejection, documentation and resolution of quality deficiencies, trend analysis, identification of corrective action, and coordination with Government inspectors. This plan will become a compliance document upon contract award and will remain in effect for the life of the contract. This plan must specify the work the Contractor must inspect and frequency of inspections (not less than daily). CAFB may require changes to the QCP during the life of the contract.

5.1.2 The Contractor is responsible for project quality control, from design through completion. The Contractor must provide and maintain a comprehensive Quality Control Program to ensure that all materials and workmanship thoroughly comply with the CDs. CAFB inspections do not constitute de facto quality control for Contractor on this project. The QCP includes:

39. Monitor suppliers, manufacturers, products, services, site conditions, and workmanship, to ensure specified construction quality.
40. Ensure all completed new construction is protected from damage, deterioration or degradation of any kind.
41. Thorough compliance with manufacturers' instructions and tolerances (if manufacturers' instructions or tolerances conflict with Contract Documents, obtain CO clarification before proceeding).
42. Application of more stringent standards, tolerances and codes, whenever two or more disagree.
43. Verify work is performed only by personnel qualified to produce the specified quality.
44. Verify field measurements exactly correspond to CDs, Shop Drawings or manufacturer instructions.
45. Verify connections and use anchors designed to withstand stresses, vibration, distortion, or disfigurement.
46. Prevent the accumulation of product fabrication and installation tolerances.
47. Monitor and adjust product dimensions and positioning before securing in place.

5.1.3 The QCP must incorporate both acceptable work quality and government inspections. The Contractor must inspect and document all work quality compared to the minimum standards. Documentation must include completion and submittal of a "Contractor's Quality Control Report (QCR) Daily Log of Construction – Military" to the CE Inspector each day. QCP standards and reference actions include:

48. Comply with related association, trade, or other consensus standard for products or workmanship, unless applicable codes are more stringent.
49. Conformance to reference standards whose issue dates precede the final Contract Documents date unless code sets a specific date.

50. Obtaining copies of standards where required by product specification sections.

51. If specified reference standards conflict with CDs, obtain CO clarification before proceeding.

52. Corrective Actions for Defects in the Work including:

- Replace the Work, or portions of the Work, not conforming to specified requirements. The Work or portions of the Work will both be referred to as “Work” or “the Work”.
- If, in the opinion of the CMI, it is not practical to remove and replace the Work, the defective Work may remain, but the CO will direct an appropriate remedy or determine a payment adjustment.
- The authority of the Government to assess the defect and determine payment adjustment(s) is final.

53. Rejected Products: Payment will not be made for rejected products for any of the following:

- Products wasted or disposed of in an unacceptable manner.
- Products determined as unacceptable before or after placement.
- Products not completely unloaded from the transporting vehicle.
- Products placed beyond the lines and levels of the required Work.
- Products remaining at work site(s) after completion of the Work.
- Loading, hauling, and disposing of rejected products.

5.1.4 Quality Control Manager QCM: The Quality Control Manager must direct Contractor’s QCP and must be responsible for plan administration and inspection of work. The QCM must during normal duty hours be available to meet with the CO upon 15 minutes notification at a location selected by the CO.

5.2 TESTING AND INSPECTION SERVICES

5.2.1 Materials Testing: If required by the Final approved Construction Documents, Contractor must employ and pay for services of an independent testing agency or laboratory acceptable to the Government. Within five (5) days of NTP issue date submit: testing laboratory name, address, and telephone number; and names of full time registered Engineer/specialist and responsible officer; and a copy of the most recent laboratory facilities inspection report of made by Materials Reference Laboratory of National Institute of Standards and Technology, with memorandum of remedies of any deficiencies reported by the inspection.

1) All tests of materials or systems must be certified and submitted in the original form.

2) Concrete compressive strength test cylinders or borings must be taken in the presence of, and at the times and locations designated by the CMI. Contractor must notify the CO 36 hours prior to any concrete placement. The CO must notify the CMI 24 hours prior to concrete placement. Contractor must immediately label test cylinders legibly with black permanent marker indicating cylinder number, project number, and date.

3) Perform compaction tests of fill material in the presence of the CMI, who will specify each test location. Compaction tests will be required for each lift. Contractor must notify the CMI at least 24 hours in advance of each test and results of any compaction test performed without the CMI present will not be accepted.

4) Inspections, tests, and reports of lab test results made by Contractor, and Contractor agents must be furnished according to the Project specifications. Contractor must bear all costs for tests required for quality assurance. Government requested tests beyond those required by the CDs, the

QCP or standard industry practice will be paid by the Government. Copies of environmental related test results must be to the CO via Submittal.

5.2.2 Testing Consultant: An independent firm will perform tests, inspections and other services specified in the Construction Documents and as required by the CO. Testing equipment must be calibrated at reasonable intervals with devices of accuracy traceable to the National Institute of Standards and Technology or to professionally accepted values of natural physical constants. Testing, inspections and source quality control may occur on or off the project site. Employment of testing agency or laboratory will not relieve Contractor of obligation to perform Work in accordance with the Construction Documents.

5.2.3 Contractor Responsibilities: Assist testing firm with incidental tasks; furnish samples of materials, design mix, equipment, tools, storage; and provide safe access. Notify CMI and testing firm 24 hours before services required. Pay testing firm for required additional tests and samples. Re-testing or re-inspection required because of non-conformance to CDs must be performed by the same independent firm as directed by the CO.

5.2.4 Agency Responsibilities: Test samples of mixes submitted by Contractor. Provide qualified personnel at site. Cooperate with CMI and Contractor in providing services. Perform specified sampling and product testing according to specified standards. Ascertain compliance of materials and mixes with requirements of Contract Documents. Promptly notify CMI and Contractor of observed irregularities or non-conformance of Work or products. Perform additional tests required by Government. Attend preconstruction and all progress meetings.

5.2.5 Agency Reports: After each test, promptly submit reports in duplicate to CO, to CMI and to Contractor. Reports must indicate observations, results of tests, and compliance or non-compliance with CDs. When requested by CMI, provide test results interpretations including: Issue date; project title and number; location within the project; inspector’s name; date and time of sampling or inspection; identification of product and specification section; type of inspection or test; test date; test results; conformance with CDs.

5.2.6 Testing Authority Limits: Agency or laboratory may not release, revoke, alter, or enlarge the Contract Documents. Agency or laboratory may not approve or accept any portion of the Work. Agency or laboratory may not assume any Contractor duties. Agency or laboratory has no authority to stop the Work.

5.3 MANUFACTURERS' FIELD SERVICES

Major equipment manufacturers must provide qualified staff personnel to observe site conditions, surface and installation conditions, workmanship quality, equipment start-up, and to test, adjust and balance equipment as necessary, and to initiate instructions when necessary. Submit observer qualifications to CO for approval at least 30 days prior to site visits. Observer must report observations, decisions or directions given to applicators or installers that are supplemental or contrary to manufacturers' written instructions.

5.4 PRODUCT REQUIREMENTS

Provide products suitable for intended use from qualified manufacturers. Provide products of each type by a single manufacturer unless specified otherwise. Do not use materials or equipment removed from existing premises, unless specifically allowed by the Contract Documents. Provide interchangeable components of the same manufacturer for components being replaced.

5.4.1 Delivery, Storage and Handling: Transport, protect, handle and store products according to manufacturer's instructions. Product seals and labels must be intact, visible and legible at all times. Inspect shipments to ensure that products are undamaged and comply with CD requirements. Keep all products away from contact with the ground. Protect products from any form of soiling, disfigurement, damage, condensation and degradation to products before and after installation. Arrange stored products to permit access for inspection.

5.4.2 Substitutions: Substitutions may be considered when an Originally Specified Product (OSP) becomes unavailable through no fault of the Contractor. Provide a substitution request in report form on Contractor company letterhead including, to the CEPM’s satisfaction, all information necessary to substantiate compliance of proposed Substitution with the CDs. The cover page of a substitution request must have been reviewed by and bear approval seal and signature of the Contractor. Substitution requests must neither have the appearance of nor be attached to any Material Submittal or Shop Drawing. Using Submittal forms or process to obtain approval for substitutions will be viewed as an attempt to defraud the Government.

5.4.3 A substitution request represents that Contractor: warrants that the proposed substitution meets or exceeds the OSP quality; will provide an equal or better warranty for the substitution than for the OSP; will make changes to other Work necessary to accommodate the substitution and that those changes will not reduce the quality of the whole or any part of the Project; will revise Construction Documents (As-Builts) as necessary; waives claims for any additional costs or time extension arising from incorporation of the substitution into the Project; will reimburse Government for review or redesign services associated with re-approval by authorities.

5.4.4 Substitution Request Procedure: Submit three copies of Substitution Request for review by the CEPM. Limit each request to one proposed Substitution. Submit Shop Drawings, Product Data, and certified test results attesting to the proposed product equivalence. Burden of proof is on Contractor. The CO will notify Contractor in writing of decision to accept or reject the request. If rejected, Contractor may propose another substitution.

5.5 CLOSEOUT PROCEDURES

5.5.1 Final Cleaning: Execute prior to final project assessment. Clean interior and exterior glass, surfaces exposed to view; remove temporary labels, stains and foreign substances, polish transparent and glossy surfaces. Clean equipment and fixtures to a sanitary condition with appropriate cleaning materials. Clean operating equipment filters. Clean debris from roofs, gutters, downspouts, and drainage systems. Clean site; sweep paved areas, rake clean landscaped surfaces. Remove waste and surplus materials, rubbish, and construction facilities from the site.

5.5.2 Starting Of Systems: Coordinate schedule for start-up of various equipment and systems. Notify Government seven days prior to start-up of each item. Verify that each piece of equipment or system has been checked for proper lubrication, drive rotation, belt tension, control sequence, and for conditions which may cause damage. Verify tests, meter readings, and specified electrical characteristics agree with those required by the equipment or system manufacturer. Verify that wiring and support components for equipment are complete and tested. Execute start-up under supervision of applicable personnel in accordance with manufacturer’s instructions. Submit a written report that equipment or system has been properly installed and is functioning correctly.

5.5.3 Demonstration and Instructions: Demonstrate operation and maintenance of products and equipment to Government's personnel two weeks prior to Substantial Completion date. A manufacturer's representative who is knowledgeable about the Project must demonstrate equipment and products. If equipment or systems require seasonal operation and Substantial Completion is out-of-season, perform demonstration within six months. Use operation and maintenance (O&M) manuals as basis for instruction. Review contents of manual with Government personnel in detail to explain all aspects of operation and maintenance. Demonstrate operation, control, adjustment, troubleshooting, servicing, and maintenance of each item of equipment at agreed time, at CO designated location. Prepare and insert additional information in operations and maintenance manuals when need for additional data becomes apparent during instruction. Provide adequate amount of time required for instruction on each item of equipment and systems necessary for complete cognizance by government maintenance personnel, but not less than indicated in individual specification sections.

5.5.4 Operation and Maintenance Manuals: Submit bound in 8-1/2 x 11 inch (A4) text pages, three D size ring binders with durable plastic covers. Binder cover(s) must bear: the title "OPERATION and MAINTENANCE INSTRUCTIONS"; project title; project number; and binder subject matter. Subdivide binder contents with reinforced punched tabbed permanent page dividers; with section dividers clearly printed under reinforced laminated plastic tabs; organized logically as described below:

54. Drawings: Bind with text; fold larger drawings to size of text pages.
55. Contents: Prepare a Table of Contents for each volume, with each product or system description identified, typed on white paper, in three parts as follows:
56. Part 1: Directory, listing names, addresses, and telephone numbers of Contractor, Subcontractors, and major equipment suppliers.
57. Part 2: O&M instructions arranged by system and subdivided by specification section. For each category, identify names, addresses, and telephone numbers of Subcontractors and suppliers. Identify: Significant design criteria; List of equipment; Parts list for each component; Operating instructions; Maintenance instructions for equipment and systems; and Maintenance instructions for special finishes, including recommended cleaning methods and materials, and special precautions identifying detrimental agents.
58. Part 3: Project documents and certificates, including the following: Shop drawings and product data; Air and water balance reports; Certificates; Originals and photocopies of warranties and bonds.

5.5.5 Submit draft copy of completed volumes 21 days prior to final inspection. This copy will be returned with Contracting Officer comments after final inspection. Revise content of all document sets as required prior to final submission. Submit two sets of revised final volumes, within 14 days after final inspection.

5.5.6 Spare Parts and Maintenance Products: Specify spare parts and extra products adequate for equipment and products maintenance for two years beyond the warranty period. Specify project site delivery location(s) and require delivery receipts prior to final payment to Contractor.

5.5.7 Product Warranties and Bonds: Require the following: Obtain and collect transferrable warranties, and bonds executed in duplicate by responsible subcontractors, suppliers, and manufacturers, within 14 days after completion of related work; Verify documents are legal, complete, and notarized; Co-Execute submittals when required; Provide Table of Contents and assemble in three D side ring binder with durable plastic cover; Submit prior to final Payment Application. Require these document submittal times: For equipment or component parts put into service during construction and with CO permission: within 14 days of acceptance; Others: within 14 days of Substantial Completion, before final Payment Application; If acceptance delayed beyond Substantial Completion: within 14 days after acceptance, stating acceptance date as the beginning of the warranty or bond period.

5.5.8 Maintenance Service: Specify that: Service and maintenance must be furnished during warranty periods; System components must be examined at a frequency consistent with reliable operation and be cleaned, adjusted and lubricated as required; Parts must be repaired or replaced whenever required; Only parts produced by the original component manufacturer must be used; and Maintenance service must not be assigned or transferred to any agent or Subcontractor without prior written consent of the CO.

End of Quality Requirements

CHAPTER 6 SAFETY, HEALTH AND ENVIRONMENTAL PROTECTION

6.1 SAFETY AND HEALTH

6.1.1 Contractor Responsibility: The Contractor bears full responsibility and liability for compliance with all Occupational Safety and Health Administration (OSHA) and applicable state and local safety and health regulations for this Project and holds the Government harmless for any Contractor action resulting in illness, injury, or death. The Contractor is solely responsible for ensuring the safety of all Contractor and non-contractor personnel whenever and wherever present on CAFB. Additional safety and health requirements stated or implied in this SOW apply to Contractor work that could affect non-contractor personnel at CAFB.

6.1.2 Applicable Publications: The publications listed below form a part of these Base Standards to the extent referenced herein or in the Project scope. The publications are referred by basic designation only. Wherever listed publications are applicable, the Contractor must comply with the latest extant publication.

1) Code of Federal Regulations (CFR)

- OSHA General industry Safety and Health Standards (29 CFR 1910) Publications
- OSHA Construction Industry Standards (29 CFR 1926)
- National Emission Standards for Hazardous Air Pollutants (40 CFR, Part61)
- Hazardous Chemical Reporting (40 CFR 370)

2) Federal Standards (Fed Std)

- 313A Safety Data Sheets, Preparation and the Submission of Safety and Health requirements, DA Circular 40-83-4

3) No asbestos containing materials will be allowed in performance of this contract or anywhere on CAFB.

6.1.3 Safety Program: The Contractor must organize, maintain and document a program to ensure compliance with safety requirements, with personnel assigned to manage its functions including: administrative, liaison, and technical to implement, direct, monitor, control, and allocate resources for all safety tasks within all aspects of the work. The Contractor must ensure full compliance of all on-site subcontractors with all safety requirements. All safety documents and related data must be available in the Contractor site office for review by the CO at any time.

6.1.4 Confined Spaces: Before entering a "Confined Space", the CMI must brief the Contractor on the space's known hazards. The Contractor must develop a Confined Space plan and obtain a Confined Space permit. Government provided information and issued permit notwithstanding, the Contractor remains fully responsible for the safety of anyone directly or indirectly affected by work in that confined space.

6.2 GENERAL

6.2.1 Contracting Officer (CO) and assigned Quality Assurance Evaluator (QAE) or Construction Inspector (CI) are terms not specified in this document. The generic CO is used throughout to designate the Government representative with contract authority. NOTE: Contacts for base offices are NOT to be included in revisions of this document. The point of all Contractor contact with the installation is the CONTRACTING OFFICER.

6.2.2 "Contractor" refers to the prime and all sub-contractors. Everyone on site must follow the agreements made with the signatory Contractor.

6.2.3 The Contractor is responsible for compliance with all federal, state, and local environmental laws and regulations, as well as Air Force regulations and guidance. These requirements include, but are not limited to, items discussed in this section.

6.2.4 This installation is subject to federal, state, and local inspections to review compliance with environmental protection laws. An inspection by the Environmental Protection Agency (EPA), Mississippi Department of Environmental Quality (MDEQ), Air Force Civil Engineer Center (AFCEC), or any other environmental or safety enforcement agency may include questioning and inspection of Contractor personnel, sites, and processes.

6.2.5 Contractors must complete and provide documentation as requested by the Contracting Officer or by inspectors, including documentation of any environmental training required by federal, state, or local regulations.

6.2.6 If a Notice of Violation (NOV) is issued to the Government or Contractor due to the Contractor's error or omission, the Contractor will be held liable for damages—including fines, penalties, or corrective actions imposed by federal, state, or local agencies.

6.2.7 If there is an accidental spill, the Contractor must follow installation spill reporting and clean up procedures as outlined in 6.8.

6.3 HAZARDOUS MATERIAL

6.3.1 Submit the following list BEFORE the project starts: Submit a list of ALL wet and dry chemical materials to be used during the performance of the contract with an estimated total annual use on site, MSN (if available), manufacturer, container size, and intended use to the CO before project start. Materials are to be checked in through (and tracked by) the 14CES/CEIE HazMat Tracking Activity (HTA) as they arrive on station with a current Safety Data Sheet specific to the manufacturer, container quantity, and total quantity brought on station. Changes of quantity or manufacturer are to be immediately cleared through the CO and the 14CES/CEIE HMMP HTA. All drums and/or containers of liquid materials must be marked to indicate contents. Unmarked containers are not allowed on site. All containers must be kept closed when not in use.

6.4 PROJECT EXECUTION

6.4.1 Hazardous materials must be stored and used in accordance with all federal, state, and local laws and regulations. Hazardous materials must not be stored in containers in direct contact with the ground. Containers must be in good condition with no holes, leaks, creases, or excessive rust. Containers must be kept closed when not in use. No hazardous materials/waste will be stored on the installation without CO approval.

6.4.2 Safety Data Sheets (SDS) must be available on site (at the location) for all hazardous materials.

6.4.3 Any hazardous or special waste generated must be handled in accordance with all local, state, and federal laws and regulations, including RCRA requirements for waste management and DOT requirements for waste transport. The Contractor must be responsible for obtaining all necessary permits, licenses, and approvals unless otherwise specified in writing by the CO.

6.5 HAZARDOUS WASTE GENERATION/DISPOSAL

6.5.1 The Contractor must maintain records of all waste determinations, including appropriate results of analysis performed, substances and sample locations, date and time of collection, and other pertinent data as required by 40 CFR Part 280, Section 74 and 40 CFR, Part 262, Subpart D.

6.5.2 The Contractor must provide a copy of all hazardous waste manifests to the CO no later than close of business the same day waste is transported off base. The Contractor must provide a closed copy of the hazardous waste manifests not later than 30 days after the waste is transported off base or before final payment, whichever occurs first. Since CAFB's EPA identification number will be used on all hazardous waste manifests, 14 CES/CEIE must sign the manifest. The Contractor still retains all liabilities associated with containerizing, loading, shipping, and disposing of the hazardous waste.

6.6 RECYCLING

6.6.1 The Contractor may elect (in the contract) to manage their own recycling of materials. In such cases, tonnage (weight slips) of material disposed of and reused or recycled (including reports of "no activity" in a given month) must be reported to the CO no later than the 5th of each month until the contract is completed. The COR must transfer all such reports to the CO upon receipt.

6.6.2 The Contractor must notify the CO a minimum of 3 business days prior to the need for recycle materials containers or changes in container size or delivery schedule so the appropriate containers and delivery schedule can be established.

6.6.3 A roll-off recycle metal materials container can be provided to the Contractor by 14 CES/CEIE upon request.

6.6.4 The following materials are acceptable for inclusion in the container provided for recycling:

- Copper: sheeting, wire, devices with an extension cord, electronics, motors.
- Steel: iron, sheet, structural beams, components.
- Aluminum: sheet, structural components, cast material and containers.
- Lead: batteries, sheeting, pipe
- Paper: cardboard, office paper, mixed paper.

6.6.5 The following items may be included in the provided containers with exceptions as noted. Again, the saving to the Contractor is the avoided disposal fees.

- Oversize materials (limbs/steel/aluminum) may not need reducing by the Contractor. Contact the recycling center directly to have exceptions inspected for handling at Government expense.
- Glass: bottles only, not mixed with construction debris collection. Office material recycling allows for the mixing of paper/glass/cans/plastic to be dropped off at the recycling center during business hours.
- Wood/scrap lumber/limbs: under 6" diameter, non-treated, nail/screw/mixed material free. Wood not meeting these criteria are construction debris.
- Plastic bottles: #1&2 only, empty of any liquid or dry material. Rinsing not required.
- Other: must be cleared with the recycling staff.

6.6.6 C&D diversion on or off base is reportable in tons to the CO. Items and exception are:

- Wood: limbs, pallets, clean straight lumber scrap may be collected for wood mulch BUT no screws, hardware, metal straps, bolts, plastic, Formica or treated products are wanted. The dumping of unwanted materials is a contract violation and the Contractor will be directed to dispose of the material properly at their expense.
- Concrete: as fill, riprap, cut and reused, or crushed.
- Asphalt: millings used for new construction.
- Gravel: used decorative gravel may be counted as a diverted material if it is suitable for another purpose. Plastic and other materials are to be removed prior to reuse as a ground cover.

6.6.7 Materials NOT meeting any of the above criteria will be considered "construction debris" and must be disposed of at the Contractor's expense.

- Foam, Styrofoam, sheet plastic contaminated with dirt.
- Furniture, chairs, office dividers, mixed material items are generally not acceptable.
- Plate glass, glass objects, table tops, Pyrex.
- Liquids are generally not acceptable. Oil may be recycled but not in the provided mixed material container.
- Wood with nails or paint/brick/masonry/sheetrock.

6.6.8 The Contractor MUST NOT recycle pressurized containers, containers with free liquid, free solids, or other material. Container puncturing, emptying, crushing, or cutting MUST be in accordance with MDEQ and installation Air, Water and disposal regulations and permit conditions.

6.7 PROHIBITED MATERIAL/ACTIONS

6.7.1 If materials not indicated in the current contract requirements are encountered that may be dangerous to human health upon disturbance at any time, such as polychlorinated biphenyl's (PCBs), lead-based paint, and friable and non-friable asbestos, the Contractor is to take no action that would disturb or further disturb the material encountered and must notify the Contracting Officer (CO), and assigned Quality Assurance Evaluator (QAE) or Construction Inspector (CI) immediately. Failure to make such notification relieves the Government from all liability for cost and/or performance time impacts until such time proper notifications are received by the CO. Verbal notification must be followed up with written notification within three workdays.

6.7.2 Class I Ozone Depleting Substances (ODS) as defined in Section 602(a) of the Clean Air Act must not be used in the performance of this contract. Radioactive materials or instruments capable of producing ionizing radiation—as well as materials that contain asbestos, mercury or PCBs—are also prohibited, as are materials that contain potentially hazardous concentrations of lead. The Consumer Product Safety Commission states that potentially hazardous concentrations of lead are greater than 90 parts per million or 0.009%. Exceptions to the use of the above excluded materials may be considered by the CO, upon written request by the contractor; however, their use is a violation of federal law, AFI, executive orders, or policy.

6.7.3 Disposal of any chemical or materials into the sanitary sewer system or out in the environment where it can enter the storm system or any natural waterway is prohibited. Disposal of any hazardous material on base is prohibited.

6.8 SPILLS AND CLEANUP

6.8.1 If there is a spill—of any amount or type—of hazardous materials or waste to include all petroleum products, the Contractor must immediately call 911 to call the installation fire dispatch providing description of material spilled (including SDS), time and location of spill, estimate of amount spilled, spill surface, and path of spill (i.e., Did the material enter or will it soon enter a drain inlet, waterway, soil, etc.?). Note that spills as used here includes releases of materials from ruptured or leaking storage receptacles (such as tanks), from transmission lines and valves, either as a part of contract work or as a result of contract work on nearby storage or transmission lines, or the leak of hazardous gases such as chlorine into the air. A follow-up written report to the CO is required within 3 business days.

6.8.2 The Contractor must take immediate action to contain and clean up spills. If the Contractor is unable to begin cleanup action within 15 minutes of the spill, the Government reserves the right to begin cleanup, continue cleanup efforts/actions until the Contractor can properly take over, or enlist the services of another contractor to begin and continue cleanup as determined by the CO.

6.8.3 All costs incurred by the Government until Contractor is capable of taking control, as determined by the on scene commander, will be the sole responsibility of the Contractor. Contractor must ensure equipment, material, and trained personnel are available to meet response time and cleanup requirements.

6.8.4 Spill cleanup personnel must be trained and certified to perform spill cleanup. The Contractor must assume responsibility for the proper characterization and disposal of any waste and cleanup materials generated. All waste and associated cleanup material must be removed from the base and transported/stored in compliance with applicable regulations until final disposal.

6.8.5 The Government reserves the right to control spill cleanup activities. The on scene commander will determine best practices for spill cleanup. The CO remains as the only Air Force representative that may amend contract requirements.

6.8.6 The Contractor is responsible for restoring a spill site to the condition of the site prior to the spill or to an improved condition.

6.9 AIR CONSIDERATIONS

6.9.1 Columbus AFB is in an air attainment zone. The base operates under a true minor permit that requires extensive data gathering to verify no violations of the permit. Fueling operations, painting other than buildings, media blasting, and dust generation require reporting to the CO. The requirements for monthly reporting, if needed, will be coordinated with the CO.

6.10 AFFIRMATIVE/GREEN PROCUREMENT/BIO-BASED/ENERGY

6.10.1 There is a series of executive orders that require the purchase of preferred materials. Whenever possible—and assuming comparable quality, availability and price—supplies are to be of the appropriate minimum energy efficiency or better, made from recovered or recycled material, or made of renewable materials. Certified materials with appropriate labeling is a good guide. The CO has fiduciary obligation to follow executive orders and enforce them on the Contractor.

6.11 ASBESTOS/LEAD BASED PAINT

6.11.1 After coordination with the CO, all asbestos manifests must be signed by 14 CES/CEIE Haz Waste manager before leaving the installation. A copy of certifications must be delivered to the CO (555 Simler BLVD, CAFB MS 39710-5010) three business days before start of work.

6.11.2 No lead based paint will be utilized by any Contractor on any base facility or equipment. Removal of paint or disturbing of surfaces may result in exposing lead based paint. All such loose material, chips, and debris will be removed and not allowed to contaminate the ground or other areas as part of the contract work. All lead based paint contaminated dust/chips must be surrendered to the CO. Government will dispose of such dust/chips at Government cost.

6.12 DRINKING WATER

6.12.1 Water main breaks must be reported immediately. The base emergency customer service system will take emergency reports and after-hours calls, use the number 662-434-2856. The contracting office may become involved with liability or repair issues.

6.12.2 Back Flow Prevention is MANDATORY on all connections to the drinking water system. Taking water directly from hydrants for contractor use is prohibited without the use of a back flow preventer. Drinking water utility use may be metered; estimated use must be agreed to in advance.

6.13 EMS

6.13.1 Contractors are required to take USAF EMS awareness training and certificates must be submitted to 14 CES/CEIE for recordkeeping. Where workers are transitory (less than 160 hrs. annually), the requirement may be waived as long as a supervisor or foreman has the training. The USAF does NOT automatically recognize ISO 7000 or 14001 training or certification as equivalent.

6.14 HISTORICAL AND CULTURAL ARTIFACTS

6.14.1 Upon discovery of human remains, pottery, or other artifacts that may or may not be of Native American origin, the work site MUST BE secured, work in the area of discovery halted, and the CO immediately notified. The Contractor should also immediately notify the CO of any damage to such artifacts. The worksite will be released to the CO (or all practical portions) by 14CES/CEIE archaeological representative. Columbus AFB has no registered historic buildings or sites. Even so, there are culturally significant attributes of the chapel, there is a known cemetery on site, and WWII debris could be discovered.

6.15 NATURAL RESOURCES

6.15.1 Preserve natural resources within project boundaries and outside limits of permanent work. Do not cut, deface, or remove trees or shrubs without CO permission unless allowed under the contract. Protect existing trees that may be damaged by contractor activities. Do not fasten ropes, cables, or guying to trees without CO approval.

6.15.2 Do not disturb fish and wildlife. Do not alter water flows or significantly disturb native habitat adjacent to the project site.

6.16 NOISE

6.16.1 Noise in excess of 85dB within 50 feet of occupied space is to be avoided, muffled, baffled, or of short duration. Occupants are to be notified in advance should long term (6hrs or more) noise be anticipated due to Contractor work.

6.17 PESTICIDES/FERTILIZERS

6.17.1 Contractors must be licensed to apply any pesticide. Coordinate and report all pesticide and fertilizer application with the CO. All quantities used/applied must be reported by the 5th of the month.

6.17.2 If pest control, termite treatment for new slabs, or any type of similar actions are required by the Contract, see Attachment 1 for additional requirements. Attachment 1 is in our templates folder as Pest Control document. Be sure to attach the document.

6.18 STORM WATER PROTECTION

6.18.1 Columbus AFB has a MDEQ Industrial Storm Water Permit. All discharges to storm drains, surface water, or indirectly to the ground are regulated and reportable. Even “clean” water is subject to the permit. The washing of vehicles will be in designated areas only. The use of soap outside of these sites is strictly prohibited. Wash water will not be dumped on the ground, parking lots, or in storm drains. Paint and paint brushes MUST NOT be rinsed outside. Such actions must be done in a sink that has a septic hookup. Options may include capture of all fluids and transfer to an agreed upon sanitary sewer dump point. Situations and specifics are open to discussion; however, fluids must not be disposed of on the ground without a MDEQ approved Industrial Storm Water Permit modification.

6.18.2 Any bare dirt or off road activity will generate sediments. These are considered a pollutant to the waters and must be controlled with dikes, silt cloth, hay bales, and other measures. Records of weekly inspections of work sites may be demanded and must be kept current. Equipment and metals will not be placed directly on the ground.

6.18.3 If the project involves open excavations, surface water diversion methods must be taken in order to minimize entrance of runoff into the excavations. If site characteristics present potential for sediment removal due to storm water, drains in the area must be protected using hay bales, silt screens, or approved equivalent.

6.19 REQUEST FOR CE SERVICE/ACTION/CLEARANCE; AF FORMS 332/813

6.19.1 These are the key forms to getting government furnished space repaired, clearing trenching operations outside of the originally identified work area, or bringing in or ending an operation. The CO will help you coordinate on these documents, if and when needed.

6.19.2 Mississippi One Call 811 must be used before the start of any dirt work.

6.20 TANKS

6.20.1 Prevent oily or other hazardous substances from entering the ground, drainage areas, or local bodies of water. Secondary containment is required around temporary fuel oil or petroleum storage tanks of any size, if any leakage could enter a storm drain, ditch or other waterway. Notify the CO prior to bringing any tanks on- site. Any container 55 gallons or greater is regulated as requiring secondary containment in the state of Mississippi. Smaller containers should be protected in a spill locker or on a spill pallet or other device that can contain 110% of the container contents.

6.20.2 Contractor tanks will meet all tank, air, and storm water requirements, be in good repair, not rusted, labeled, sheltered, and be removed when no longer needed.

6.21 LIMITS AND EXCEPTIONS

6.21.1 General: These Contractor guidelines apply to all Contractors working on Columbus Air Force Base, except those transient in the nature of their tasks (such as delivery operations that have no other presence on the base).

6.21.2 EMS Training: Services of a transitory nature may be exempt from EMS training. Qualifying transitory contracts have very limited presence on the installation, making EMS training impractical. A carpet cleaning service working completely out of a service truck, preforming a single short term task, having no on site presence would qualify as exempt. A lawn service storing fuel overnight would not. The CO is to include all Contractors unless there is a compelling reason not to. Non-resident delivery and freight services are exempt (such as Fed-ex).

6.22 LEGAL AUTHORITY

6.22.1 Contractors will follow the listed clauses. This guide has given local information pertaining to these and is not to be considered complete information. The Contractor is instructed to become familiar with the full text of each clause as they have specific applications.

- Safe Drinking Water Act http://water.epa.gov/lawsregs/rulesregs/sdwa/index.cfm
- Clean Water Act http://www2.epa.gov/laws-regulations/summary-clean-water-act
- Clean Air Act http://epa.gov/oar/caa/caaa_overview.html
- Resource Conservation and Recovery Act (RCRA) http://www.epa.gov/agriculture/lrca.html
- Comprehensive Environmental Response, Compensation and Liability Act http://www.epa.gov/superfund/policy/cercla.htm
- Code of Federal Regulations (CPR) series 29 (OSHA), 40 (Environmental) and 49 (DOT)

End of Safety, Health and Environmental Protection